COLLEGE OF ENGINEERING AND ARCHITECTURE

KUVENTORY: Kape Uno Inventory Management System

In Partial Fulfillment of the Requirement in

Software Design

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2. INTRODUCTION

2. Background of the Study

In today's highly competitive food service industry, effective inventory management has become a critical factor in ensuring operational efficiency, minimizing costs, and maintaining customer satisfaction. Restaurants depend on the continuous availability of ingredients and supplies to provide quality products and services. However, many small and medium-sized food establishments still rely on manual, paper-based inventory systems that are often inefficient, labor-intensive, and susceptible to human error. These traditional methods frequently lead to inaccurate inventory records, delayed stock replenishment, excessive food wastage, and financial losses caused by poor inventory control. Chen et al. (2023) emphasized that effective inventory management significantly improves business efficiency by reducing inventory-related costs, ensuring product availability, and supporting better operational performance. Likewise, Bah et al. (2023) found that proper inventory management positively influences an organization's profitability by minimizing inventory losses and improving resource utilization. As digital technologies continue to transform business operations, organizations are increasingly adopting information systems to automate inventory processes, improve data accuracy, and support more effective managerial decision-making (Laudon & Laudon, 2022).

In the Philippines, the adoption of digital technologies among micro, small, and medium enterprises (MSMEs) remains a growing priority despite the continued use of conventional inventory management practices. Many local restaurants continue to maintain handwritten inventory logs because they are inexpensive and familiar to business owners. Nevertheless, manual inventory management often results in stock discrepancies, delayed updates, misplaced records, and computational errors that contribute to overstocking, stock shortages, food spoilage, and unnecessary operational expenses. Recognizing these challenges, the Department of Trade and Industry (DTI, 2023) encourages MSMEs to embrace digital transformation by adopting technology-driven solutions that improve productivity,

operational efficiency, and long-term business competitiveness. This initiative highlights the importance of implementing digital inventory systems to enhance business operations among local enterprises.

Recent advancements in web-based inventory management systems have provided businesses with more effective tools for monitoring and controlling inventory operations. Unlike traditional methods, these systems allow real-time recording of inventory transactions, centralized data storage, automated stock updates, and the generation of inventory reports that support strategic planning and purchasing decisions. Erameh and Odoh (2021) developed a web-based inventory control system for small and medium enterprises that integrates QR codes and barcodes to monitor inventory movement, provide real-time inventory information, reduce inventory processing time, and improve operational efficiency.

Similarly, Oladele et al. (2021) developed an inventory management system using association rule techniques that enhanced inventory monitoring, data organization, and decision-making. These studies demonstrate that web-based inventory systems can significantly improve inventory accuracy and operational performance compared to traditional paper-based methods. In addition, Laudon and Laudon (2022) explained that integrated information systems optimize business processes, improve resource utilization, and provide timely information that supports managerial decision-making.

Despite the documented benefits of digital inventory management systems, the specific inventory practices of small food-service establishments may still involve manual recording and monitoring processes that are not adequately addressed by general-purpose inventory solutions. In the case of Kape Uno, inventory activities involve the monitoring of stocks across Bodega, Kiosk, Portion, and Grilled areas, where stock quantities, movements, and other inventory information must be recorded and monitored regularly. The continued reliance on manual records creates difficulties in maintaining accurate and timely inventory information, monitoring stock movement, tracking expiring items, and

identifying items that require replenishment. These conditions indicate the need for a system specifically designed around the inventory workflow of Kape Uno. In response, KUVENTORY: Kape Uno Inventory Management System is proposed as a web-based solution intended to centralize inventory records, support stock monitoring and movement tracking, facilitate FEFO-based inventory handling, provide inventory-related notifications, and generate organized inventory reports. The system is intended to address the identified operational gaps while providing a more centralized and systematic approach to inventory management.

2.2. Statement of the Problem

The manual inventory management at Kape Uno is fraught with operational inefficiencies and constant human errors in managing raw stocks, materials, kitchen equipment, and supplies in four different stations (Bodega, Kiosk, Portion, and Grilled). The inefficiencies result from the absence of real-time information on stocks due to the use of manual log books, lack of coordination in inter-station stock transfers, lack of monitoring of spoilage of perishables due to failure in implementing FEFO tracking, and late stocking of inventory due to the absence of automated low inventory alerts. Human errors include accounting errors in manual entries, human errors in calculations in shift changes, loss through spills, damages, and misplacement of physical logs.

Specifically, this study seeks to answer the following questions:

1. How should the web-based inventory management system (KUVENTORY) be created and implemented in place of manual record-keeping by Kape Uno to automate daily stock counting?
2. How effective is the proposed system as evaluated by the proponents and admins based on the ISO/IEC 25010 software quality standards in terms of:

   2.1. Functional Suitability

2.2. Performance Efficiency

2.3. Compatibility

2.4. Usability

2.5. Reliability

2.6. Security

2.7. Maintainability

2.8. Portability

3. How should the system facilitate automatic notifications on low stock and impending expiration dates of supplies to avoid down time and product shortage?
4. How reliable and efficient is the performance of the KUVENTORY software when doing real-time stock reduction and batch adjustment operations amidst high volume of kitchen activities?

2.2. Objectives of the Study

2.2.1General Objectives

This research aims to design, develop, and evaluate the effectiveness of KUVENTORY: Kape Uno Inventory Management System, a web-based inventory management system with real-time stock alerts for Kape Uno Bistro located in Tuguegarao City, Cagayan. The system is proposed to enhance inventory monitoring and stock control by providing faster item identification, real-time stock updates, and timely low-stock.

2.2.2 Specific Objectives

The study specifically aims to:

- Design and develop a secure web-based inventory management system for monitoring and managing inventory records.
- Develop a low-stock alert notification system wherever the stock levels of ingredients or supplies fall below the required threshold, thereby supporting timely restocking and preventing inventory shortages.

- Generate records that will provide summaries of stock levels, inventory movement, and restocking needs.
- Evaluate the developed system based on ISO/IEC 25010 tool.

Significance of the Study

For the Owner of Kape Uno Restaurant. This study provides a web-based inventory management system that improves inventory accuracy, enables real-time stock monitoring, supports timely replenishment, and enhances operational efficiency.

For the Food Service Industry. This study highlights the benefits of adopting web-based inventory management systems to improve inventory control, reduce operational inefficiencies, and support better resource management.

For Software Developers. This study provides insights into the development of web-based inventory systems using QR code technology, automated inventory recording, and real-time stock monitoring, which may serve as a reference for future system enhancements.

For Future Researchers. This study serves as a reference for future research on web-based inventory management systems and digital transformation in the food service industry.

Definition of Terms

FEFO (First-Expired, First-Out) - Inventory management approach/algorithm embedded in the system that encourages the use or issuance of stock items that have the earliest expiration date in order to minimize the occurrence of perishable food waste and spoilage.

Inventory Management System - Software that is employed to monitor, control, and manage stock levels, ingredient consumption, reorder points, and stock movements in an establishment.

Kape Uno - The local bistro/restaurant found in Carig Sur, Tuguegarao City, Cagayan, which becomes the main subject, research site, and implementation location of the system.

KUVENTORY—Web-based inventory management system customized for Kape Uno to automate stock monitoring, batch management, FEFO expiration tracking, and low-stock reminders.

Low-Stock Alert System - Automated notification system that notifies the owner or system administrator when ingredient/supply stocks fall below a certain level.

Real-Time Stock Reduction - Automatic deduction of ingredient level in the database based on the transaction/kitchen consumption.

Stock-In - The process or activity of stocking newly arrived ingredients and supplies.

Stock-Out - The process or activity of withdrawing or de-stocking ingredients and supplies from the system due to kitchen consumption, wastage, or spoilage.

Scope and Limitations

This study aims to design, develop, and evaluate the effectiveness of KUVENTORY: Kape Uno Inventory Management System as a web-based inventory management platform for Kape Uno Restaurant located in Carig Sur, Tuguegarao City, Cagayan. The scope encompasses real-time stock monitoring across four stations (Bodega, Kiosk, Portion, Grilled), daily worksheet reconciliation, FEFO batch expiration tracking, stock movement audit trails, automated notifications, immutable report snapshots, and role-based access control.

This study is limited to the inventory management operations of Kape Uno Restaurant. Out of scope for this system are public client account registrations, online food ordering, Point of Sale (POS) cashiering engines, supplier purchase order portals, accounting ledgers, employee payroll/HR, and CRM functions.

3. CONCEPTUAL FRAMEWORK

INPUT	PROCESS	OUTPUT
● Authentication Log-in Details ● Stock Details ● Items - Description - Unit Cost ● Supplier ● Quantity Input ● Quantity Balance	● Authentication User Credentials ● Sort, Search, Update, Stock Details ● Archive Data	● **KUVENTORY: Kape Uno Inventory Management System** ● Generation of Reports

Figure 3.1 Conceptual Framework

The Kape Uno Inventory Management System (KUVENTORY) will be using the Input-Process-Output (IPO) approach. To make sure that the system will meet the operational needs of the business, it will be imperative for the proponents to collect all the required functional information through an interview conducted on the owner and employees of Kape Uno Restaurant.

The Input stage encompasses all the raw data obtained and kept in the main KUVENTORY database. In this stage there is Owner Information which helps ensure system security as it holds usernames, encrypted passwords, and account recovery information. The input includes active Inventory Data which holds master inventory list, inventory movements, inventory manual adjustments, low stock parameters, and full log of stock-in and stock-out movements. Finally, the transactions data is included in this stage such as Purchases (purchase order number, purchase date, suppliers, product list, total cost of purchase, and status - Pending or Delivered) and Sales and Transactions (transaction number, customer information, quantity of items, total cost of transaction, and status - Completed or Not Completed).

The Process stage involves automation of the system operations, business logic, and algorithms that run using the database entries to facilitate the operation process and to reduce frequently committed human errors during manual operations, such as miscalculating physical stock balances during shift

turnovers, miscounting remaining raw ingredients, failing to record stock transfers between stations, forgetting to log damaged or spilled raw materials, and misplacing paper receipts for newly delivered supplies. The system verifies the credentials for logging in and provides Role-Based Access Control (RBAC) to ensure the security of administration tasks and enables the owner and his team to enter, modify, and manage products' data. The system uses barcodes, as well as search algorithms, to identify the products quickly while checking the inventory. It processes the real-time stock-in and stock-out operations, adjusts the inventory balances of all inventory locations (Bodega, Kiosk, Portion, and Grilled), controls stock movement during the intensive kitchen activities. Furthermore, the system follows the principle of First Expired, First Out (FEFO), whereby priority is given to the use of raw material or product with the closest expiry date. This technique minimizes the risk of spoilage, ensures the quality of raw material and prevents wastage of food by ensuring that raw material with the least time left before expiry is issued before the newly acquired one (Analisis Penerapan Metode FIFO dan FEFO pada Food & Beverage Store Prama Sanur Beach Bali Hotel, 2026).

The Output Stage refers to the practical results gained from implementing the system. The key output would be the completed KUVENTORY Management System, which is a safe, fast-responding and web-based system, created especially for Kape Uno. Using the integrated dashboard and stock monitoring interfaces, multiple inventory transfers, automatic alert log, sales financials analysis and reports (PDF, Excel, CSV), the system provides full operational transparency. In the end, this helps achieve higher inventory precision, operational efficiency, stock replenishing, accurate accounting and decreased amount of perishable food waste.

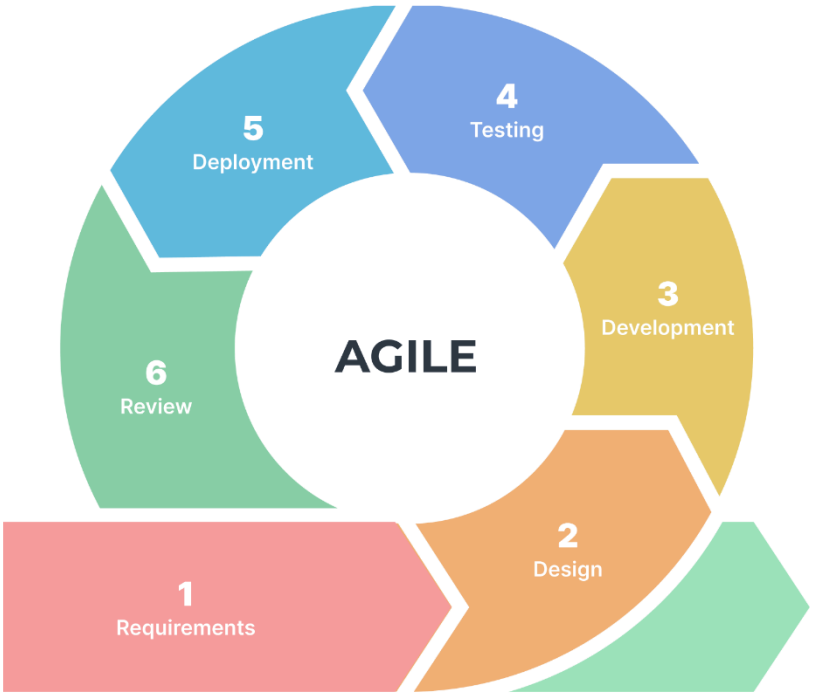


Figure 3.2 Agile Model

The implementation of the "KUVENTORY: Kape Uno Inventory Management System" involves the use of the Agile Development Model within its conceptual framework. Compared to linear models, the Agile development model follows an iterative, collaborative, and incremental approach. It enables the research and development team to continually receive feedback from the owner and employees of Kape Uno Restaurant and make adjustments to the system's features through several development cycles (or sprints). The following paragraphs elaborate the primary Agile Phases and the execution blocks of the KUVENTORY project.

At the design stage, the collected requirements will be converted into technical design and user interface (UI/UX) design. The development team will prepare the architectural diagram of the system, database schema, data flow diagram, and wireframe. At this stage of planning, the emphasis is on the design of modular and user-friendly interfaces, which will work with various screen sizes (smartphones, tablets, and laptops), and database design.

Next to the design phase, the project goes to the development phase. In the development phase, the system is developed as per the specifications set out in the design phase. Software modules are developed as per the sprint planning set out. Both front-end and back-end logics are developed through sprints. These include authentication, product management, manual stocking in and out, expense tracking, and reporting among other features developed in an incremental manner.

Upon development of a module or feature set, it moves onto the testing stage, where verification is done. This is the stage where the software that has been completed goes through a detailed process of being evaluated for any possible errors, bugs, or inconsistencies. Unit testing of the software's functions (for example, the mathematical calculations for stock reductions and FEFO) is carried out first. Next, integration testing of the entire data flow of the modules is done.

After the testing process and successful verification, the deployment phase comes into place. During this phase, the final product or prototype would be deployed to the live testing environment, which may include a web server or Netlify. The owner of the Kape Uno and key employees will be educated about the Supplies features in the first deployment. Training will be provided to employees for their daily tasks in stock logging, stock transfer approval, and report exportation. In addition, deployment makes sure that all the necessary tasks, such as FEFO Expiration Tracking, Automatic Stock Calculation, Shift Calculations (opening & closing), and Low Stock Detection, are done by the product. Deployment is defined as the official release of the product to its users.

The Review Stage is the final stage of the Agile development framework. After the deployment process, the team continues to monitor the performance of the system and addresses any issues or updates that may come up later on. In every sprint, the process involves analyzing user feedback and reviewing the functionalities of the system in the actual restaurant environment. Issues such as bug fixes, user experience improvement, and additional features are determined from user feedback collected

by the owner and other workers while carrying out normal restaurant business activities. These findings are fed back into the stock and supplies backlog until the fully developed system is acquired.

For the summary, KUVENTORY development follows an Agile model approach consisting of six phases: Requirements collects operational requirements from the owner of Kape Uno and their employees in a stock and supplies backlog; Design translates those requirements into database schemas and responsive UI wireframes; Development creates core functionalities of the system iteratively via sprints; Testing checks for accurate data integration and mathematical calculations throughout the system; Deployment puts the application to live on the Web and provides user training; and Review & Maintenance collects operational feedback from restaurants for constant improvement.

4. METHODOLOGY

The development of KUVENTORY: Kape Uno Inventory Management System followed a systematic and iterative approach to address the inventory management requirements of Kape Uno Restaurant in Carig Sur, Tuguegarao City, Cagayan. The project focused on improving the restaurant's manual inventory processes through a centralized web-based system for recording, monitoring, and managing inventory information. The methodology covered requirements gathering, system analysis and design, development, testing, deployment, training, and user evaluation.

4.1 Research/Project Approach

The project began by examining the existing inventory practices of Kape Uno to identify operational requirements, workflow concerns, and limitations associated with manual inventory management. Information gathered from actual inventory activities and existing records was used to determine the system requirements and establish the functions needed for KUVENTORY.

An Agile-oriented iterative approach was used in the development of the system. The approach allowed the development team to progressively design, implement, test, and refine the system based on identified requirements and issues. System components, database structures, interfaces, and workflows were reviewed throughout development to ensure consistency with the actual inventory processes of Kape Uno.

The overall project process consisted of requirements gathering, requirements analysis, system design, system development, testing, deployment, training, and user evaluation. This approach supported the continuous refinement of KUVENTORY while maintaining focus on inventory monitoring, stock management, expiration control, reporting, security, and usability.

4.2 Requirements Gathering and Information Procedures

To determine the operational requirements and existing inventory practices of Kape Uno, the

development team used three complementary information-gathering procedures: direct operational observation, semi-structured stakeholder interviews, and document analysis. These procedures were used to understand the actual workflow, identify operational difficulties, determine user requirements, and establish the data and functional requirements of KUVENTORY.

4.2.1 Direct Operational Observation

Direct operational observation was conducted to understand how inventory-related activities were performed within the restaurant. The observation covered actual inventory activities such as opening preparation at approximately 8:00 AM, afternoon shift handovers at approximately 1:00 PM, and closing inventory counts at approximately 5:00 PM.

The observation focused on how staff recorded and checked inventory, monitored stock levels, handled stock additions and deductions, and maintained inventory information across the Bodega, Kiosk, Portion, and Grilled operational areas. The development team also considered practical usability requirements, including the need for clear interface elements and sufficiently large touch targets when the system is accessed using tablets or other touch-enabled devices.

4.2.2 Semi-Structured Stakeholder Interviews

Semi-structured interviews were conducted with relevant Kape Uno personnel to gather information about existing inventory procedures, operational requirements, and difficulties encountered in manual record-keeping. The discussions covered inventory monitoring, shift handovers, stock handling, reporting requirements, expiration concerns, notification needs, and user access requirements.

The semi-structured format allowed the development team to follow established discussion topics while allowing additional concerns and recommendations to be documented as they arose. The information obtained from the interviews was used to refine the functional requirements and system workflows of KUVENTORY.

4.2.3 Document Analysis

Document analysis was conducted by reviewing available inventory-related records and existing documentation used by Kape Uno. These materials provided information regarding the formats and types of data recorded during inventory activities, as well as the requirements for daily inventory, stock monitoring, and historical reporting.

The findings from the document analysis were compared with the information obtained through observation and interviews. The combined findings served as references for the design of the database structure, inventory workflows, user interface, and reporting functions of KUVENTORY.

4.3 Scope of Work

The development work focused on creating an inventory management system that centralizes inventory information and supports the day-to-day monitoring and management of stock at Kape Uno Restaurant. The system covers the recording, monitoring, adjustment, tracking, and reporting of inventory-related information.

The scope of KUVENTORY includes major functional areas such as user authentication and role-based access control, dashboard overview, inventory item and category management, stock additions and adjustments, stock batch management, stock movement history tracking, daily inventory recording, FEFO-based stock consumption, low-stock and expiration notifications, inventory reporting, archived historical records, and administrative settings.

The project is limited to inventory management and does not include purchasing, sales, point-of-sale operations, cashiering, online ordering, supplier management, customer management, accounting, expense management, payroll, human resource management, customer relationship management, or other broader enterprise resource planning functions.

4.4 Software Development Methodology

KUVENTORY was developed using an Agile-oriented iterative methodology. The methodology provided a structured process for transforming identified requirements into a functional inventory management system while allowing adjustments to be made throughout development.

4.4.1 Requirements Analysis

The requirements analysis stage involved reviewing the information collected through observation, interviews, and document analysis. The development team identified the functional, data, usability, security, and operational requirements of the system.

Particular attention was given to the processes for managing inventory items and categories, recording stock additions and deductions, monitoring stock batches, conducting Daily Inventory, applying FEFO, generating notifications, producing reports, and controlling user access. The results of the analysis served as the basis for subsequent system design and implementation.

4.4.2 System Design

During the system design stage, the overall system architecture, database structure, user interface, access controls, and inventory workflows were established. The design was based on the requirements identified during the requirements analysis process.

The database structure supports inventory items, categories, stock batches, stock movements, Daily Inventory records, reports and report snapshots, notifications, audit-related records, and user or profile information. The design also incorporates FEFO-based stock management to prioritize eligible stock according to expiration dates.

The user interface was designed around the major functions of KUVENTORY, including the Dashboard, Daily Inventory, inventory management, stock monitoring, notifications, reports, and administrative functions. Role-based access was incorporated to restrict functions according to the

permissions assigned to users.

4.4.3 System Development

System development was conducted iteratively, with major system functions implemented, reviewed, tested, and refined throughout the development process. Development activities included authentication, dashboard functions, inventory item and category management, stock management, stock movement tracking, Daily Inventory, stock batch management, FEFO processing, notifications, reports, archive and historical records, user management, and system settings.

The frontend was developed as a responsive web application to support desktop, tablet, and mobile screen sizes. Backend and database functionality was implemented using Supabase and PostgreSQL, allowing inventory operations, authentication, access control, database processing, and data storage to be handled through the cloud-based system architecture.

4.4.4 Testing

Testing was conducted throughout the development process rather than only after all system components had been completed. Module-level and functional testing were performed to identify errors at an early stage, while integration-related testing was conducted as different system components were connected.

Testing focused on inventory calculations, stock transactions, Daily Inventory processes, FEFO batch processing, notifications, database operations, authentication, authorization, reporting, and system usability. Issues identified during testing were analyzed, corrected, and retested as part of the iterative development process.

4.4.5 Deployment

After the major development and testing activities were completed, KUVENTORY was prepared for deployment. The frontend application was deployed through Netlify, while Supabase Cloud provided

the backend services and PostgreSQL database infrastructure.

Deployment activities included preparing the production build, configuring the application environment, verifying database connectivity, checking authentication and access-control settings, and confirming that the major system functions operated correctly in the deployed environment.

4.4.6 Review and Maintenance

Review and maintenance activities were incorporated to support the continued operation and improvement of KUVENTORY. System issues identified during testing, deployment, and actual use were reviewed and addressed through appropriate corrections and updates.

Maintenance also includes reviewing system dependencies, database operations, security configurations, inventory calculations, notifications, reports, and other critical functions to ensure that the system remains functional, secure, reliable, and suitable for the operational requirements of Kape Uno.

4.5 Team Staffing and Resource Allocation

The project team assigned responsibilities according to the major requirements of system analysis, design, development, quality assurance, security, documentation, deployment, and user support.

Responsibilities covered system architecture and full-stack development, user interface and frontend development, quality assurance and security testing, systems analysis, documentation, training materials, deployment, and stakeholder coordination. The responsibilities were coordinated throughout the development cycle to ensure that system requirements, implementation, testing, documentation, and evaluation remained aligned.

Resource allocation was focused on the activities required to complete KUVENTORY, including requirements gathering, database and interface design, software development, testing, documentation,

deployment, training, and user evaluation.

4.6 System Development Activities

The major activities carried out during the development of KUVENTORY were organized into sequential phases. Development began with requirements gathering through observation, interviews, and document analysis, followed by the identification and analysis of inventory workflows. System architecture, database schema, and user interfaces were subsequently designed to establish the structural framework. Implementation covered user authentication, role-based access control, inventory item and category management, stock transactions, Daily Inventory functions, stock batch management, FEFO logic, notifications, reporting, and database security mechanisms. The phase concluded with thorough functional, integration, database, security, and usability testing, followed by bug fixes, deployment verification, and user orientation.

4.7 Testing Procedures

Testing was performed to verify that KUVENTORY operated according to its defined requirements and supported the intended inventory workflows.

4.7.1 Functional Testing

The major functions of KUVENTORY were tested individually to verify their expected behavior. Testing covered authentication, dashboard information, inventory items, categories, stock management, Daily Inventory, notifications, reports, archive functions, and administrator functions.

4.7.2 Inventory and Database Testing

Inventory calculations and database operations were tested to verify that stock information was properly recorded, updated, and retrieved. Stock additions, deductions, adjustments, stock batches, stock movements, Daily Inventory records, and historical information were checked for consistency.

4.7.3 FEFO Batch Verification

The FEFO function was tested to verify that eligible inventory batches were consumed according to the earliest-expiration-first principle. The stock consumption process was handled through database-level logic, with the corresponding stock movement records generated as part of the transaction.

4.7.4 Daily Inventory Testing

Daily Inventory functions were tested to verify the recording and calculation of beginning quantities, additions, deductions, and ending quantities. Draft records were checked to ensure that they did not immediately alter live inventory, while finalized inventory records were verified against the corresponding stock processing and historical records.

4.7.5 Notification Testing

Notification functions were tested under low-stock, out-of-stock, and expiration-related conditions. The system behavior for generating and maintaining notification records was checked according to the implemented database logic.

4.7.6 Authentication, Authorization, and Security Testing

Authentication and role-based access controls were tested to verify that users could access only the system functions permitted by their assigned roles. Database-level security mechanisms, including Row-Level Security and protected database operations, were also checked as part of the security testing process.

4.7.7 Usability and Compatibility Testing

The user interface was evaluated for clarity, navigation, responsiveness, and ease of use across supported devices and screen sizes. Testing considered the practical requirements of restaurant personnel who may access the system while performing inventory activities.

4.8 Training Procedures

Before full operational use, Kape Uno personnel were introduced to the functions and proper use of KUVENTORY. Training focused on helping users understand the inventory workflow within the system and the functions available according to their assigned access level.

The orientation covered system login, navigation, inventory item management, stock management, Daily Inventory, notifications, reports, and other relevant administrative or operational functions. Users were also guided on the importance of entering accurate inventory information and following the established procedures when recording stock activities.

Training and orientation were intended to support proper adoption of the system and reduce errors resulting from unfamiliarity with the new inventory workflow.

4.9 User Evaluation Procedures

User evaluation was conducted to gather feedback regarding the functionality, usability, reliability, performance, security, and overall quality of KUVENTORY. Intended users were given opportunities to interact with the completed system and perform representative inventory-related tasks.

The evaluation was aligned with the applicable quality characteristics of the ISO/IEC 25010 software quality model. The characteristics considered were Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

Quality Characteristic	Evaluation Focus
Functional Suitability	Ability of the system to provide the required inventory functions correctly and completely
Performance Efficiency	Responsiveness and processing efficiency during system use
Compatibility	Operation across supported devices, browsers, and environments
Usability	Ease of learning, navigation, understanding, and performing tasks

Quality Characteristic	Evaluation Focus
Reliability	Stability and consistency of system operations
Security	Authentication, authorization, access control, and protection of system data
Maintainability	Ease of reviewing, correcting, updating, and improving the system
Portability	Ability of the system to operate across supported environments

4.10 Development Tools and Technologies

KUVENTORY was developed using a modern web-based and cloud-supported technology stack.

Technology / Tool	Purpose
React 19	Development of the web application's user interface
TypeScript	Structured and type-safe application development
Vite	Frontend development and build tooling
Tailwind CSS	Responsive interface styling
shadcn/ui	Reusable interface components
Lucide React	Interface icons
Recharts	Data visualization and reporting charts
Supabase	Cloud backend services, authentication, and database integration
PostgreSQL	Relational database and inventory data management
Row-Level Security (RLS)	Database-level access control
PostgreSQL Functions / RPCs	Secure execution of inventory-related database operations
Database Triggers	Automated database processes and notification-related operations
Netlify	Frontend deployment

The combination of these technologies supports the development of a responsive, cloud-based inventory management system while providing mechanisms for authentication, database security, inventory processing, reporting, and system maintenance.

4.11 Deployment and Implementation Procedures

The deployment and implementation process involved preparing the completed system for operational use, configuring the production environment, verifying the backend and database services, and checking the functionality of the deployed application.

The frontend application was deployed through Netlify, while Supabase Cloud provided the backend services and PostgreSQL database infrastructure. Authentication, database policies, database functions, and other security configurations were verified before operational use.

After deployment, the major functions of KUVENTORY were checked to ensure that users could authenticate, access functions according to their assigned roles, record and monitor inventory, process stock activities, manage batches, receive notifications, generate reports, and access historical information.

User orientation and training were conducted as part of implementation to familiarize Kape Uno personnel with the deployed system. Feedback and issues identified during implementation were documented and considered for correction, refinement, and subsequent maintenance.

5. EXISTING SYSTEM

5.1. COMPANY BACKGROUND

The company Kape Uno operates in Carig Sur, Tuguegarao City, Cagayan. Running on a daily basis between 8:00 AM to 5:00 PM, this company started operating with an aim of providing black coffee, unique beverages, grilled foods, and other food products to the students, employees, and local community. Apart from running normal dining services, Kape Uno has created a remarkable reputation as a locally known meeting place and dining spot where local bands, artists, and personalities come together to have fun. Furthermore, this company has become a meeting point for organizing private events, social functions, and parties.

The operations in this establishment are managed by Mrs. Feli Frogoso, whereby the various stations in the operation process include a store (Bodega), primary sales stand, a portioning station, and a grilled goods station. The management of the inventory process within the different operations stations is managed by certain members of staff, particularly by Ms. Abie and Ms. Rociel. In addition to offering on-site dining, this company receives event bookings from clients who visit either physically or through Facebook or messenger accounts.

5.2. SYSTEM DESCRIPTION

In its current status, inventory control, sales, and shift management in Kape Uno are primarily conducted using manual and paper-based systems. Opening and closing balances of the stocks per day, shift turnover (which include the Morning and the Afternoon shifts), internal stock transfers (Bodega to Kiosk, Bodega to Grilled, Bodega to Portion), procurement invoice from suppliers, and operating expenses are manually recorded using written logs and paper receipts.

Throughout the course of business since its establishment, inventory monitoring and record keeping for stocks have been done manually. At the time of their operation, Ms. Abie and Ms. Rociel manually record the inventory in-take, disbursement of stocks per shift, sales, and the end-of-the-day inventory count. In addition, essential evaluations of the inventory such as the low stock level, expiration dates of ingredients, and inventory discrepancy are periodically evaluated visually and manually.

However, the use of manual record keeping processes brings about several challenges to the process. The manual record keeping process faces challenges such as wear and tear of physical records, loss of pages, and complete loss of data which poses a risk to the reliability of inventory data over time. Looking at past stock records and conducting audits on historical supplier transactions will require a lot of work in the manual process especially during busy periods and booking of events.

In conclusion, changing from the manual process to the use of an automated inventory management process will play a key role in improving operations at Kape Uno. A centralized online platform will improve data accuracy and security. Real time stock calculations and monitoring will aid in better decision making and reduce perishable product losses.

5.3. DATA FLOW DIAGRAM

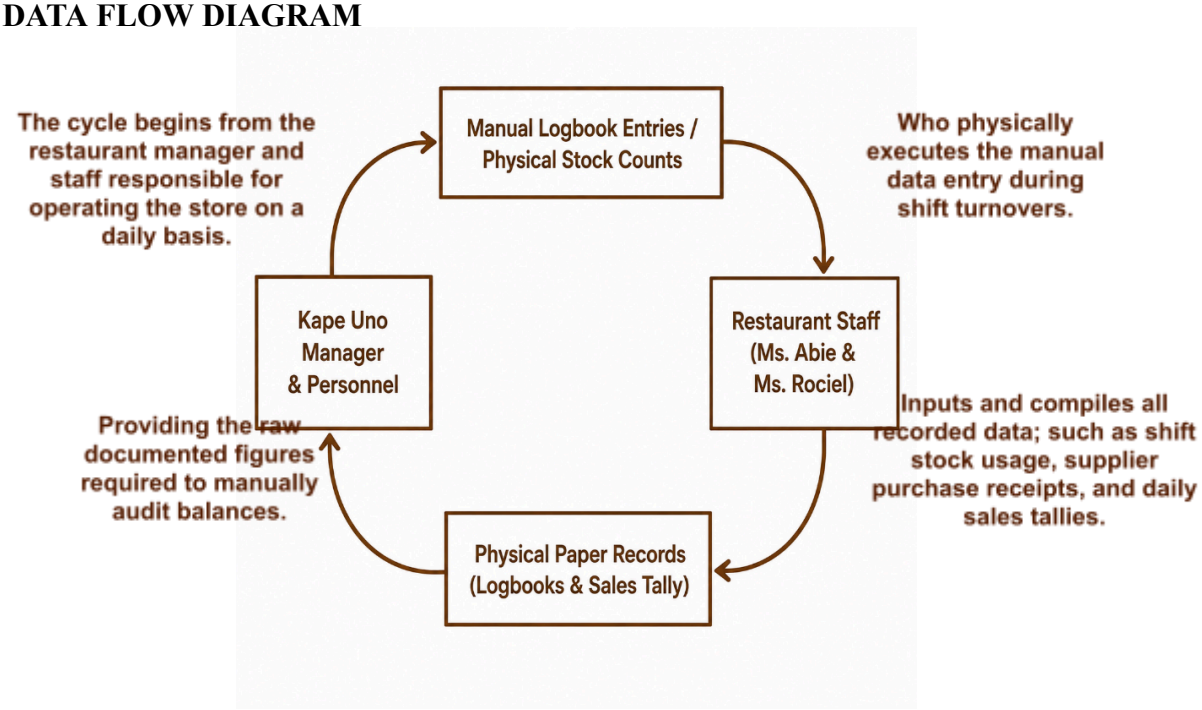


Figure 5.1. Data Flow Diagram of Kape Uno Bistro Existing System

The operational process flow of the current data workflow goes through the following process flow:

- The managers and operational staff of the restaurant physically record stock and supplies quantities, sales data, purchasing costs, and event bookings.
- The designated inventory people (Ms. Abie and Ms. Rociel) manually record their morning and afternoon stock usage in physical logs.
- The restaurant manager (Mrs. Frogoso) physically looks at sales receipts and summary sheets in order to compute their closing stock.
- The historical records of stocks, transactions, and operations are physically kept in paper folders and logbooks.

5.4. DATA DICTIONARY

Staff Record: Includes employee names, contact numbers, assigned positions, and shift assignments (morning/afternoon) documented in employee roster sheets.

Stocks Information: Comprises item codes, supplies description, categorization, unit price, inventory levels, and operational status (Active/Low Stock/Out of Stock).

Inventory Levels Log: Includes a log of physical inventory level counts at various operational points (Bodega, Kiosk, Grilled, Portion), opening stock levels, additional stock counts, and closing inventory levels.

Stock Transfer Record: Includes handwriting logs of physical inventory transfers from the central Bodega to operational stations (Kiosk, Grilled, Portion).

Procurement Receipt: Includes supplier name, dates, supply order, procurement cost, and delivery status (Pending/Delivered) documented in the physical procurement receipts.

Sales and Revenue Log: Includes channel of payment (cash, GCash, Maya, card), applied discounts (senior citizen, PWD, employee), total revenue amount, and cash on hand per shift.

Operational Expenses Log: Handwritten log of daily operational expenses (fuel, utilities, packaging materials, supplier invoices, employee meals).

5.5. PROBLEM AREAS

Based on the interview conducted with the manager and the employees of Kape Uno, the following issues have been identified by the proponents:

(1) Time-consuming manual entry process of data into the logbook, which results in human calculation errors.

(2) Challenges in managing the increasing volume of data and loss of documents: it is difficult to keep track of data using paper records, resulting in loss of documents and taxes.

(3) No real-time tracking of inventory levels and expired foods, and this may lead to unexpected stockouts or loss of food.

(4) No security features to protect sensitive information; no access control mechanism exists in case of paper records to protect financial information.

6. PROPOSED SYSTEM

KUVENTORY is a web-based application designed to consolidate the inventory management and internal operational workflows of Kape Uno. Data privacy and security are maintained through role-based access control initialized via pre-configured developer credentials, ensuring that registered users access system components strictly aligned with their designated permissions. Meanwhile, the inventory department uses the interfaces to record the daily stock balance, monitoring the transfer of raw materials from one station to another and updating the ingredient usage. The clients are allowed to create personal accounts to monitor the availability of the products and view the menu.

User and Access Management

System access operates on a role-based framework utilizing pre-set credentials established during the development phase to authenticate user accounts. Following registration and authentication, users are granted secure access restricted to permissions pertinent to their operational scope. This access control mechanism safeguards data confidentiality, protects operational integrity against security vulnerabilities, and ensures full auditability by attributing every system entry to the responsible user account.

Inventory and Stock Management

The inventory module facilitates real-time tracking of raw materials, kitchen equipment, and operational supplies across designated operational stations, including the Bodega, Kiosk, Portion, and

Grilled stations. This module manages stock transfers between stations, generates automated low-stock alerts, and logs ingredient consumption per operational shift. Furthermore, stock rotation is regulated through the implementation of the First-Expired, First-Out (FEFO) principle to monitor and mitigate ingredient expiration risks.

Data and Security Management

The platform employs data encryption protocols, secure session management, and a web-based firewall to protect critical information from unauthorized access and security threats. Automated database backups utilizing PostgreSQL are executed routinely to preserve inventory records and system logs, ensuring high data availability and robust recovery capabilities in the event of system interruptions.

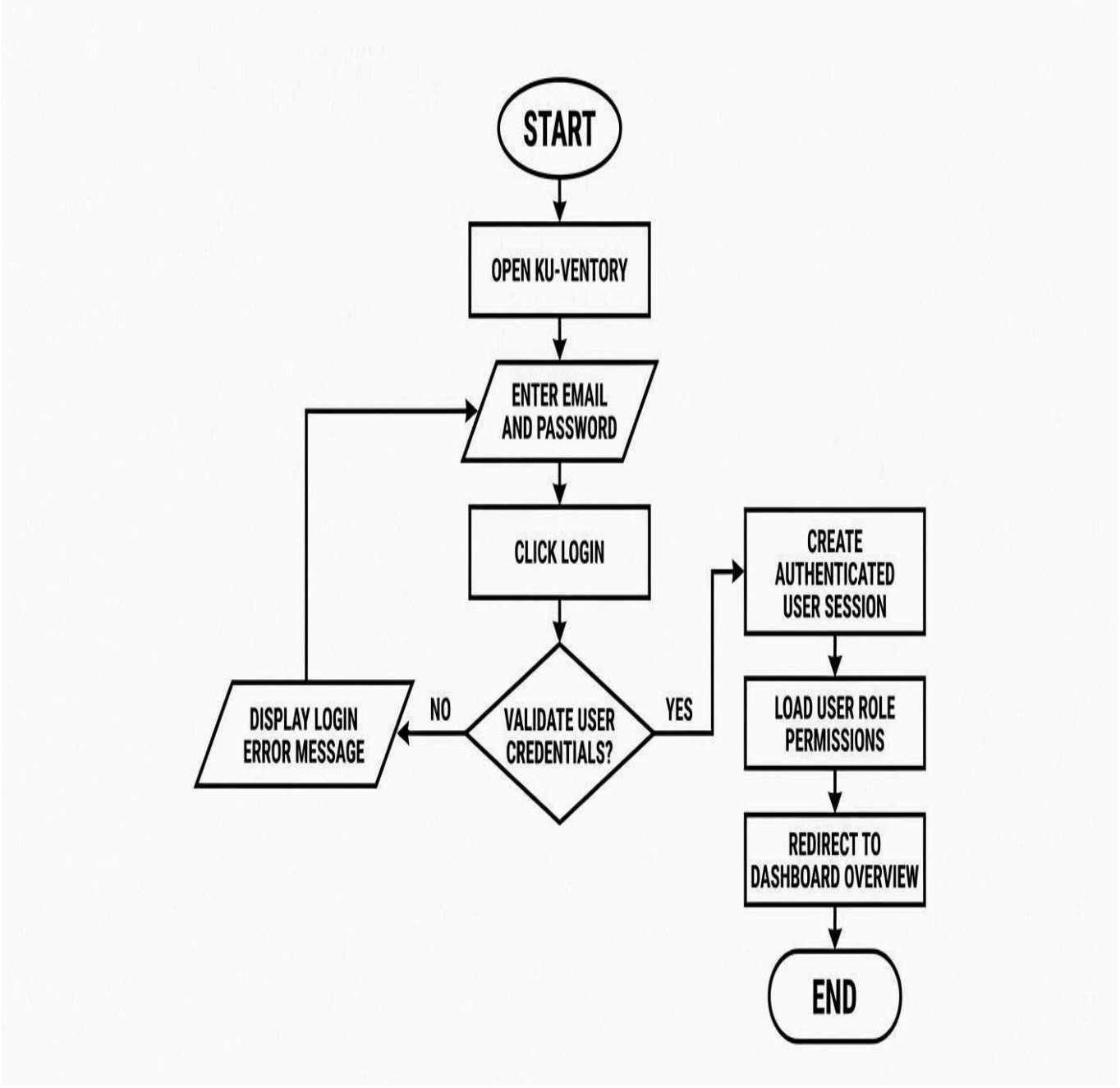
SYSTEM OVERVIEW

The KUVENTORY is a web-based inventory and management platform developed to digitize, centralize, and automate the operational processes of Kape Uno Restaurant. Instead of relying on handwritten logbooks and manual monitoring, the system provides real-time tracking of raw materials, cooking utensils, and other operational inventory across the restaurant's four major departments: Bodega, Sales Kiosk, Portioning, and Grilling. The User and Access Management module controls system access through preconfigured login credentials and Role-Based Access Control (RBAC), ensuring that users can only access the functions assigned to their roles.

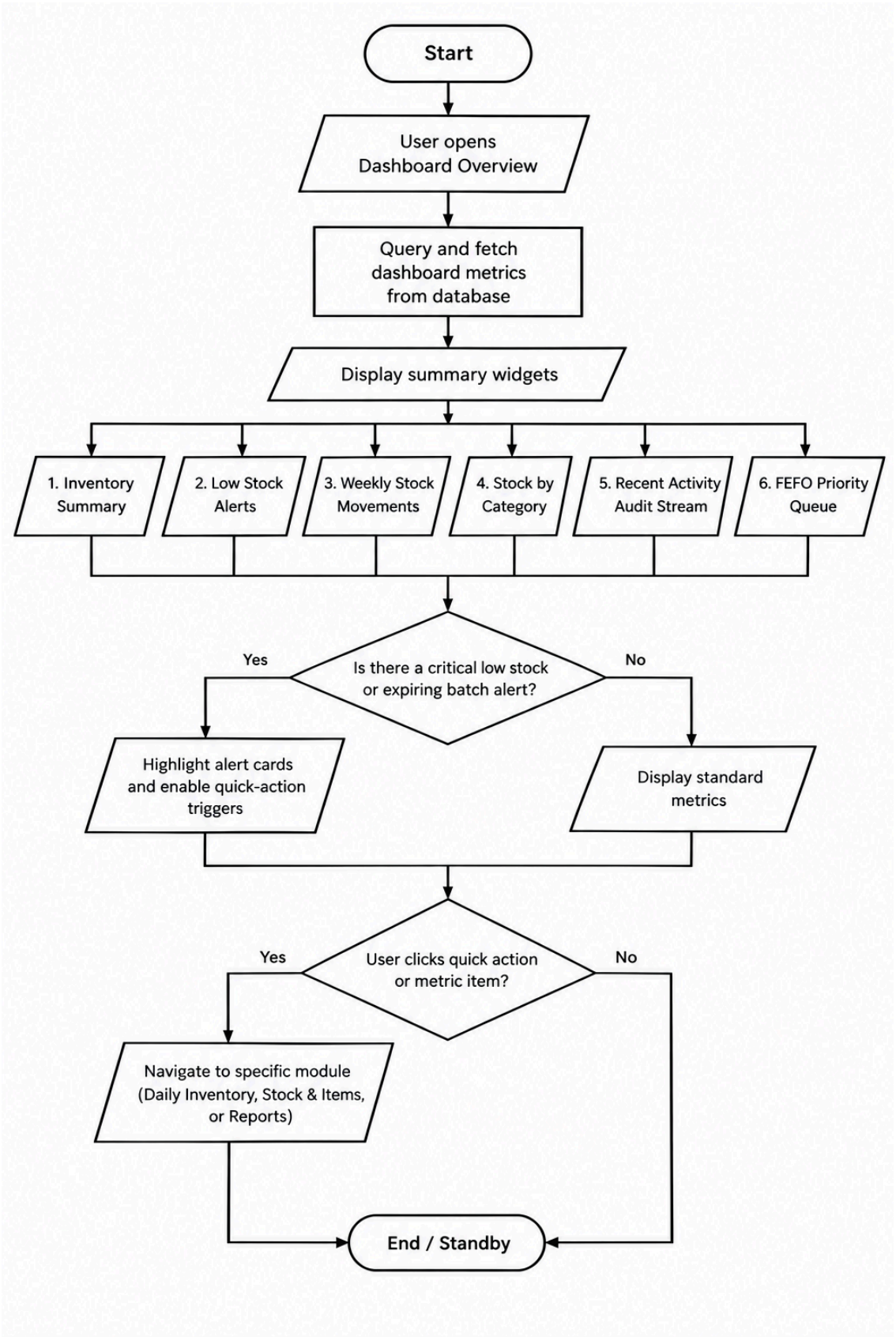
The Inventory and Stock Management module monitors the usage of raw materials per shift, manages inventory transfers between departments, generates automatic low-stock alerts, and monitors ingredient expiration dates using the First-Expired, First-Out (FEFO) method. The system also provides organized records of stock movements, allowing authorized users to monitor inventory levels and track the movement of materials between departments. Lastly, Data and Security Management ensures that inventory records and other system data are properly protected, maintained, and accessible only to

authorized users.

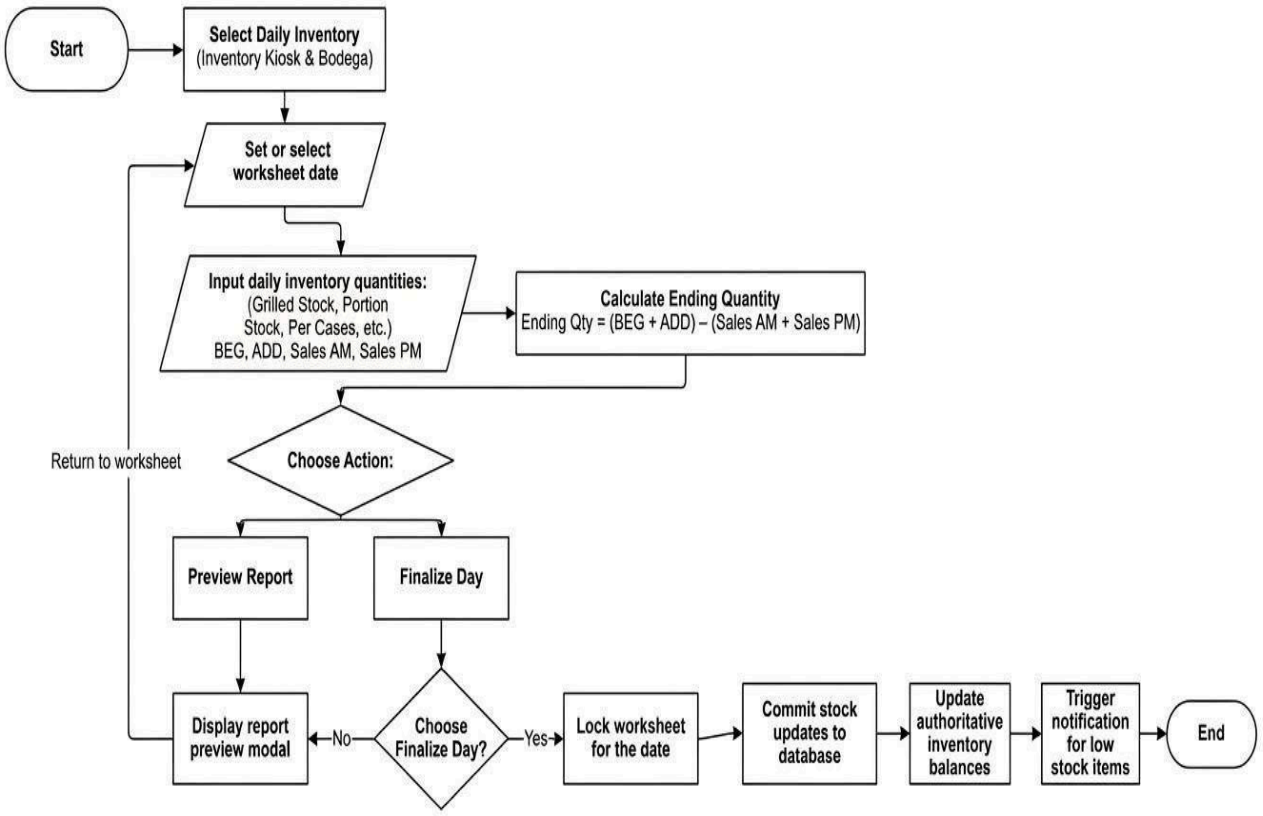
6.1.1Authentication Log in Details



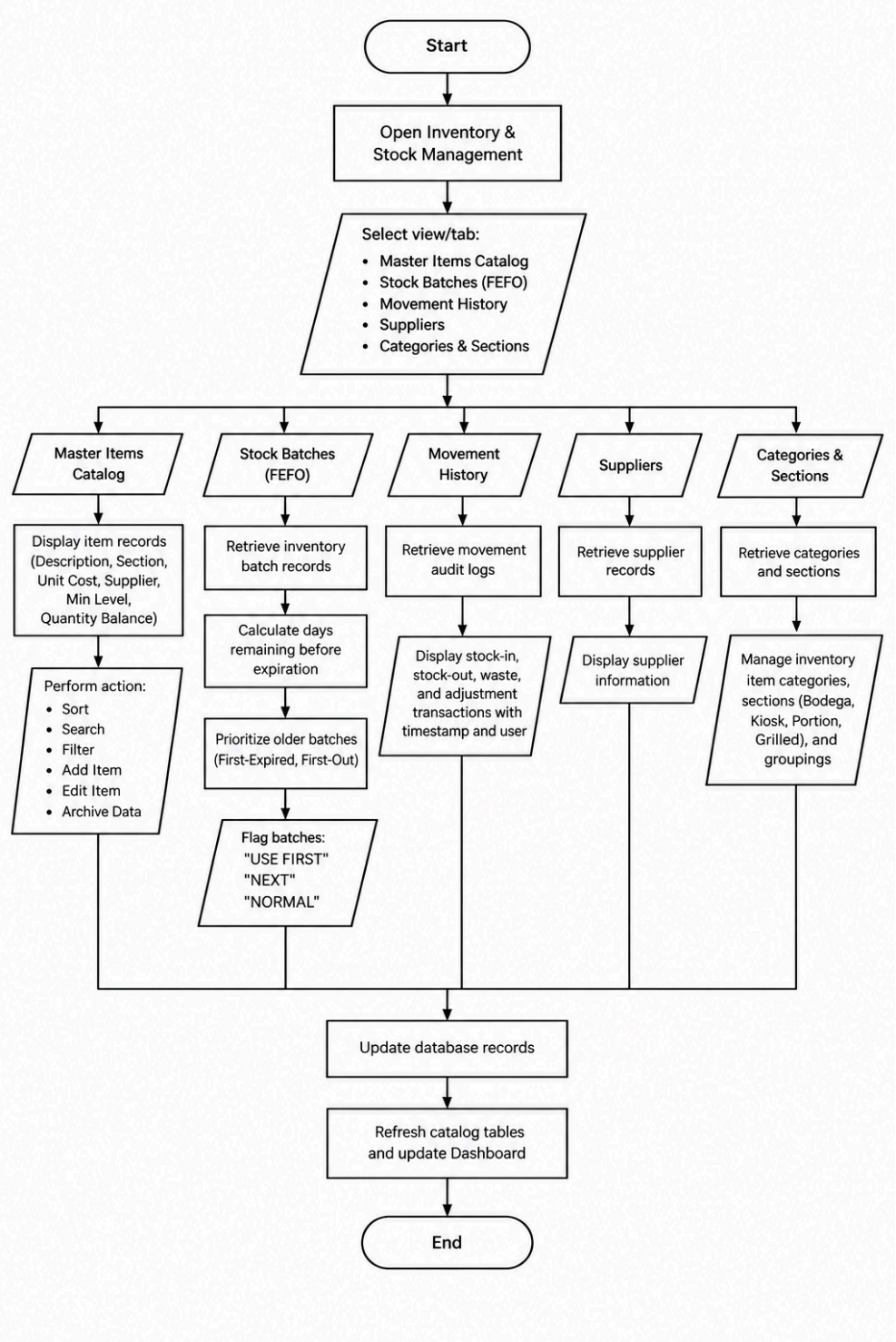
6.1.2 Dashboard Overview Module



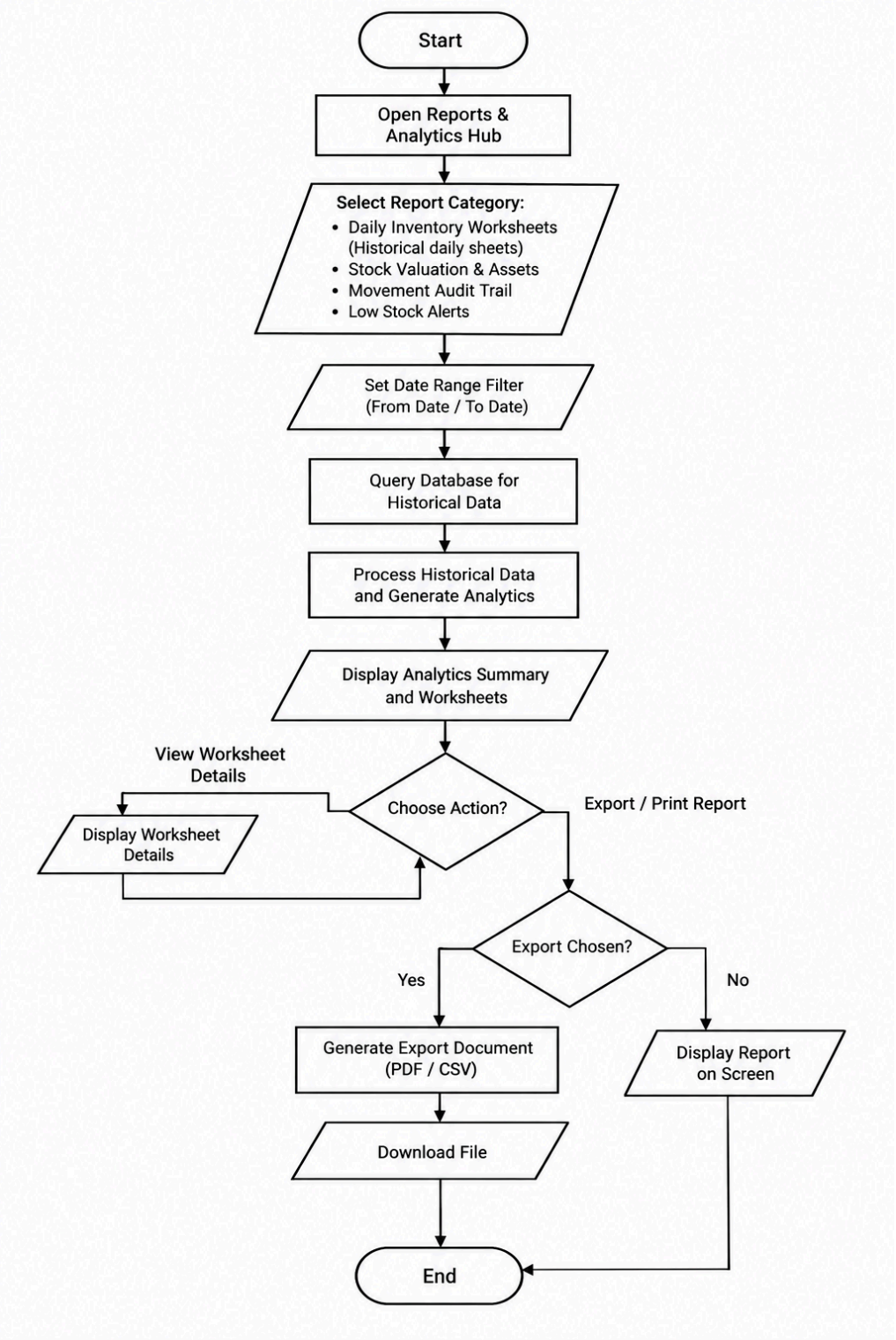
6.1.3 Inventory Kiosk and Bodega



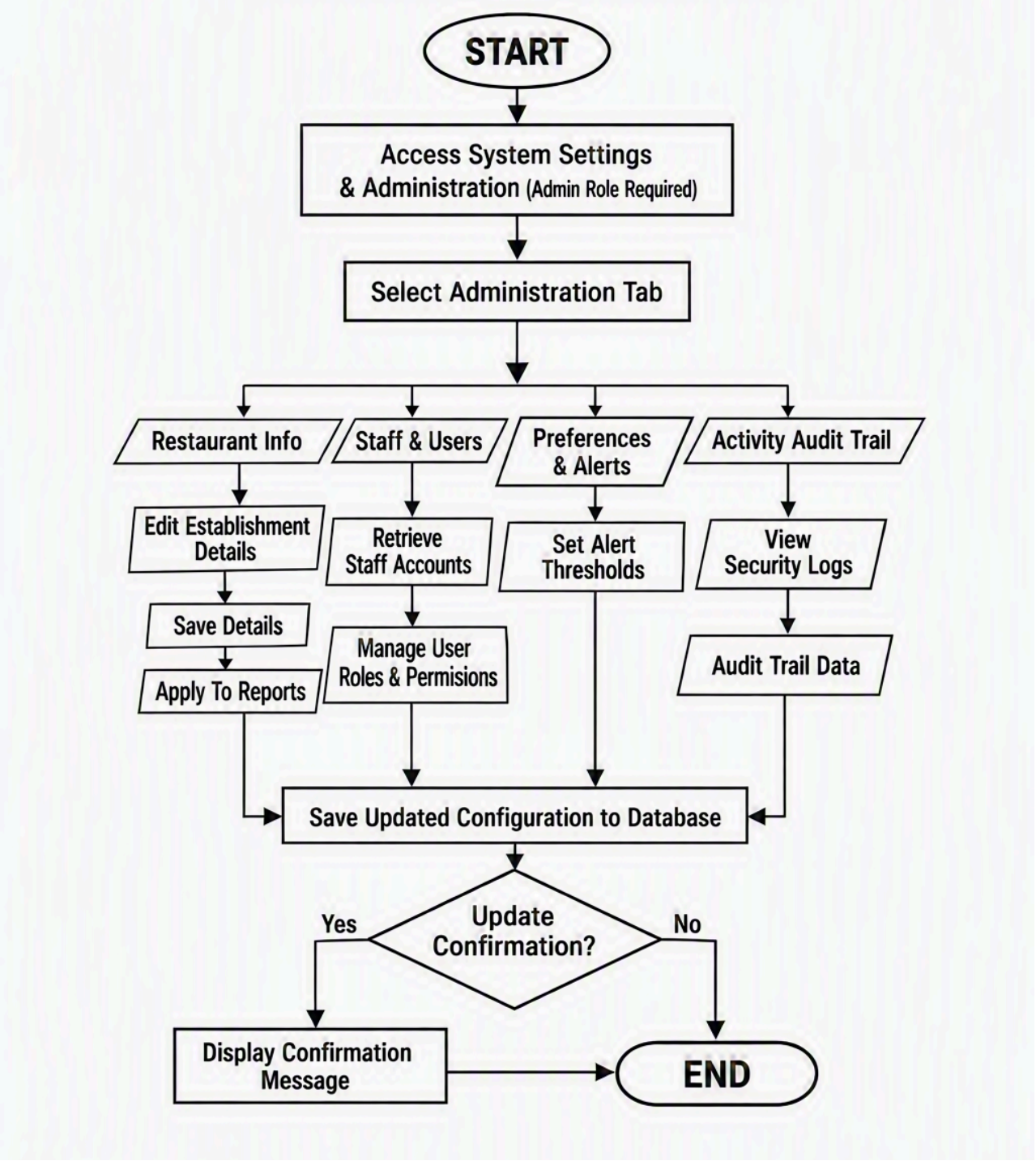
6.1.4 Inventory & Stock Management Module



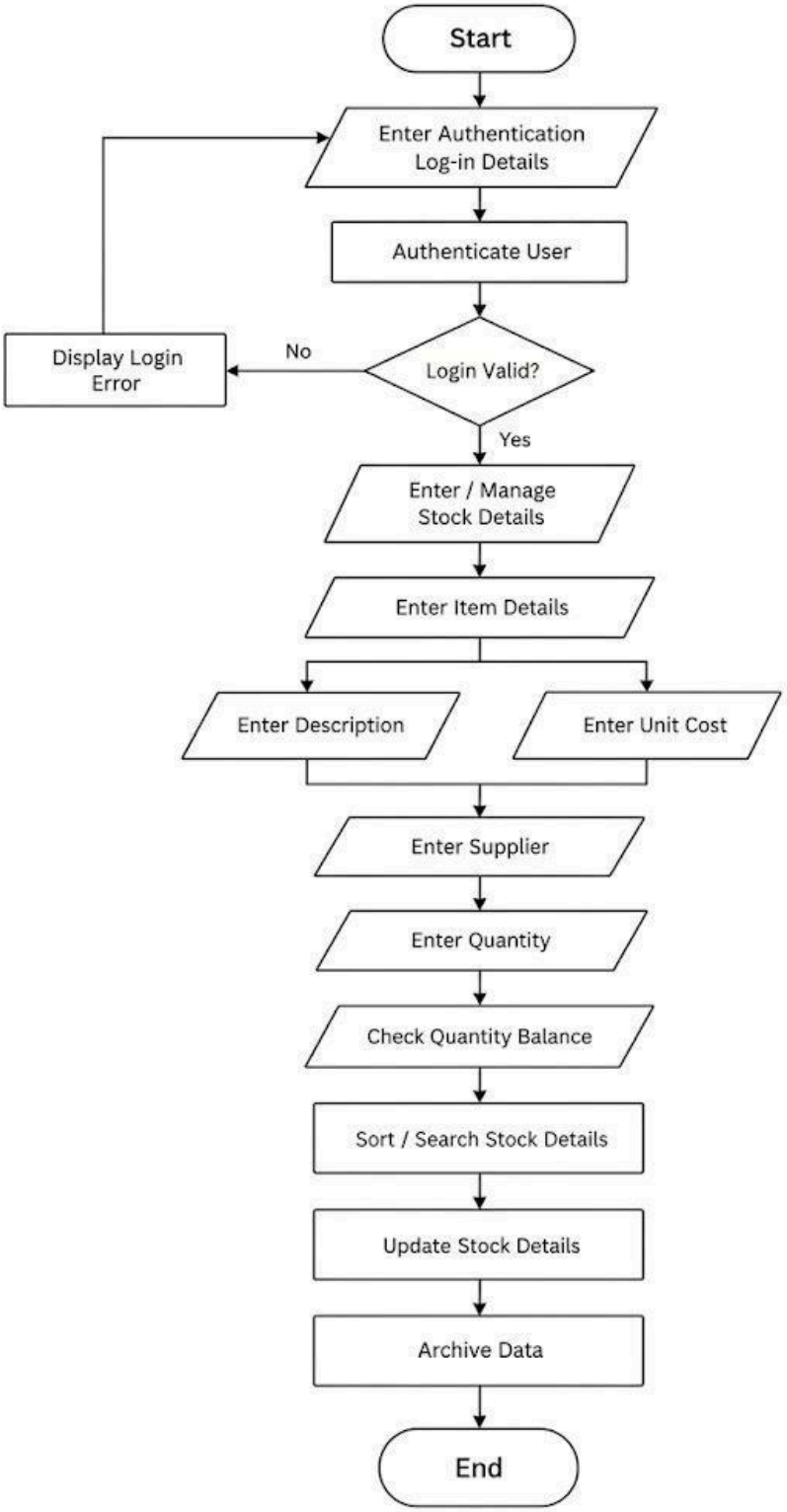
6.1.5 Reports & Analytics Hub Module



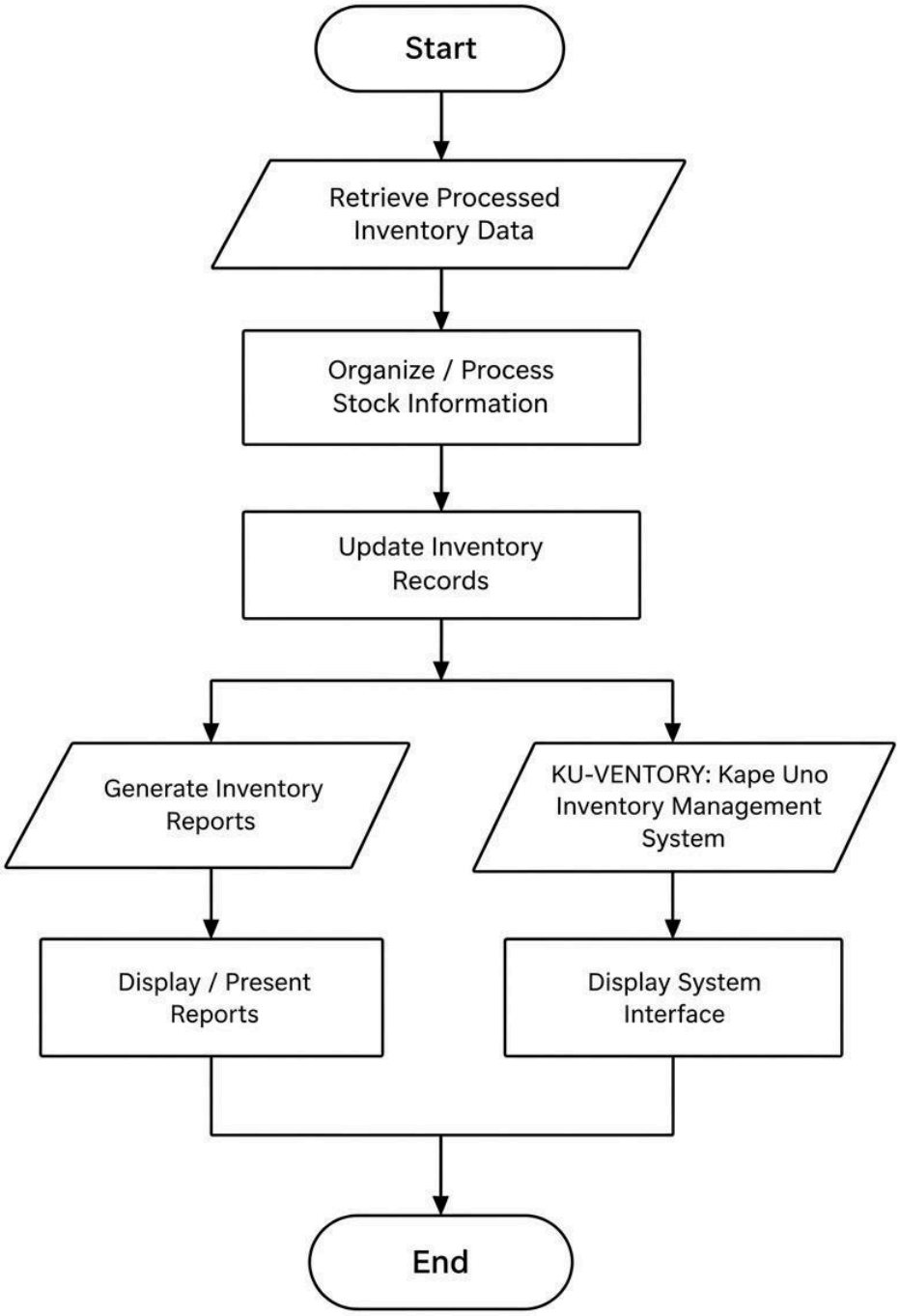
6.1.6 Users & Roles / System Settings & Administration Module



6.1.7 Input and Inventory Data


Start
Enter Authentication Log-in Details
Authenticate User
Login Valid?
No
Display Login Error
Yes
Enter / Manage Stock Details
Enter Item Details
Enter Description
Enter Unit Cost
Enter Supplier
Enter Quantity
Check Quantity Balance
Sort / Search Stock Details
Update Stock Details
Archive Data
End

6.1.8 Inventory processing and System Output



DATA DICTIONARY

A. General

- Admin / Manager – A person who manages the website of the restaurant, system settings, user roles, and operational reports.
- Inventory Staff – A person who manages shift inventory, station stock transfers, sales logging, and operational expenses (*eg, Ms. Abie, Ms. Rociel*).
- Product – Menu items, ingredients, beverages, and raw materials managed within the system.
- Station – Designated operational sectors within Kape Uno (*Bodega, Kiosk, Portion, Grilled*).

B. User Registration Input

- First Name – The first name of the user registering for system access.
- Last Name – The last name of the user registering for system access.
- Email – The email account of the user registering for system access.
- Phone Number – The phone number of the user.
- Address – The physical address of the user.
- Password – The password chosen by the user for account access.

C. Product & Stock Input

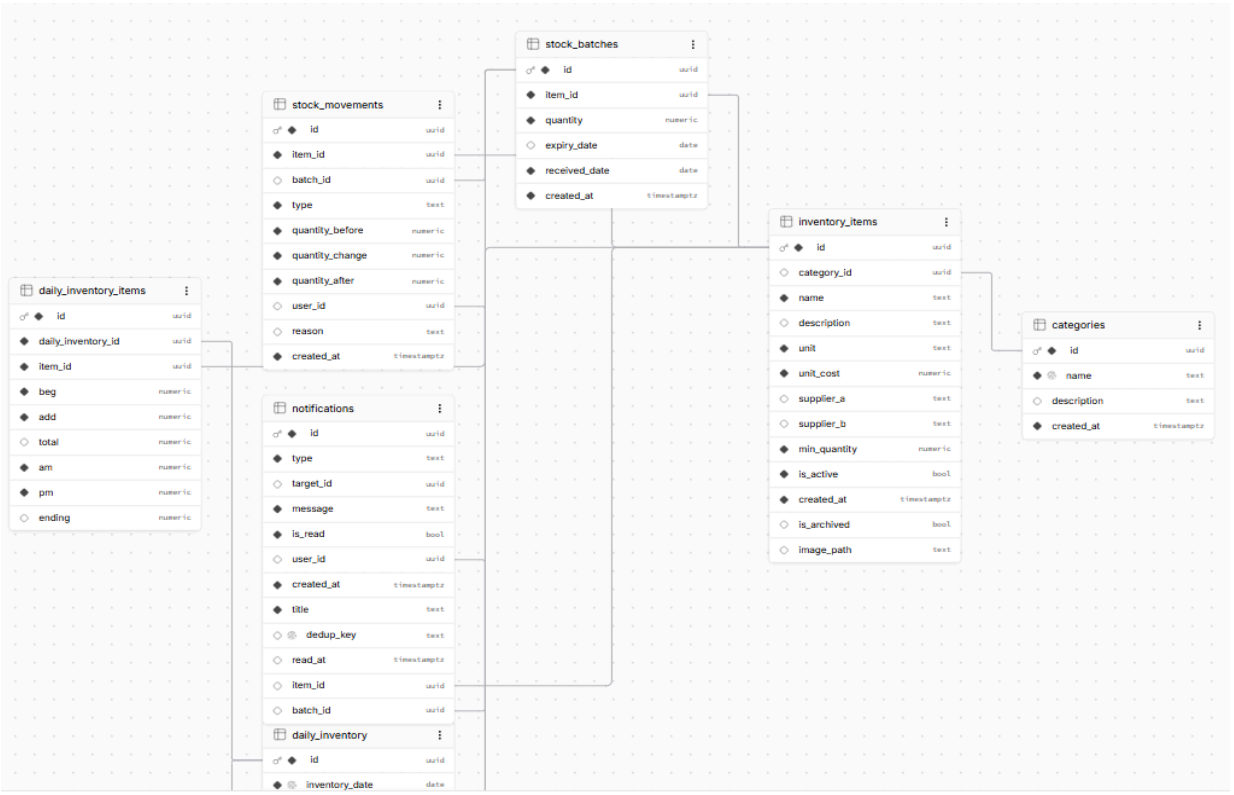
- Product Name – The official name of the ingredient, beverage, or menu item.
- Category – The functional classification of the item (*eg, Raw Ingredients, Packaging*).
- Unit Cost / Price – The purchase cost or selling price per unit.
- Minimum Stock Level – The minimum quantity threshold triggering a low-stock alert.
- Expiration Date – The shelf-life limit used for First-Expired, First-Out (FEFO) tracking.

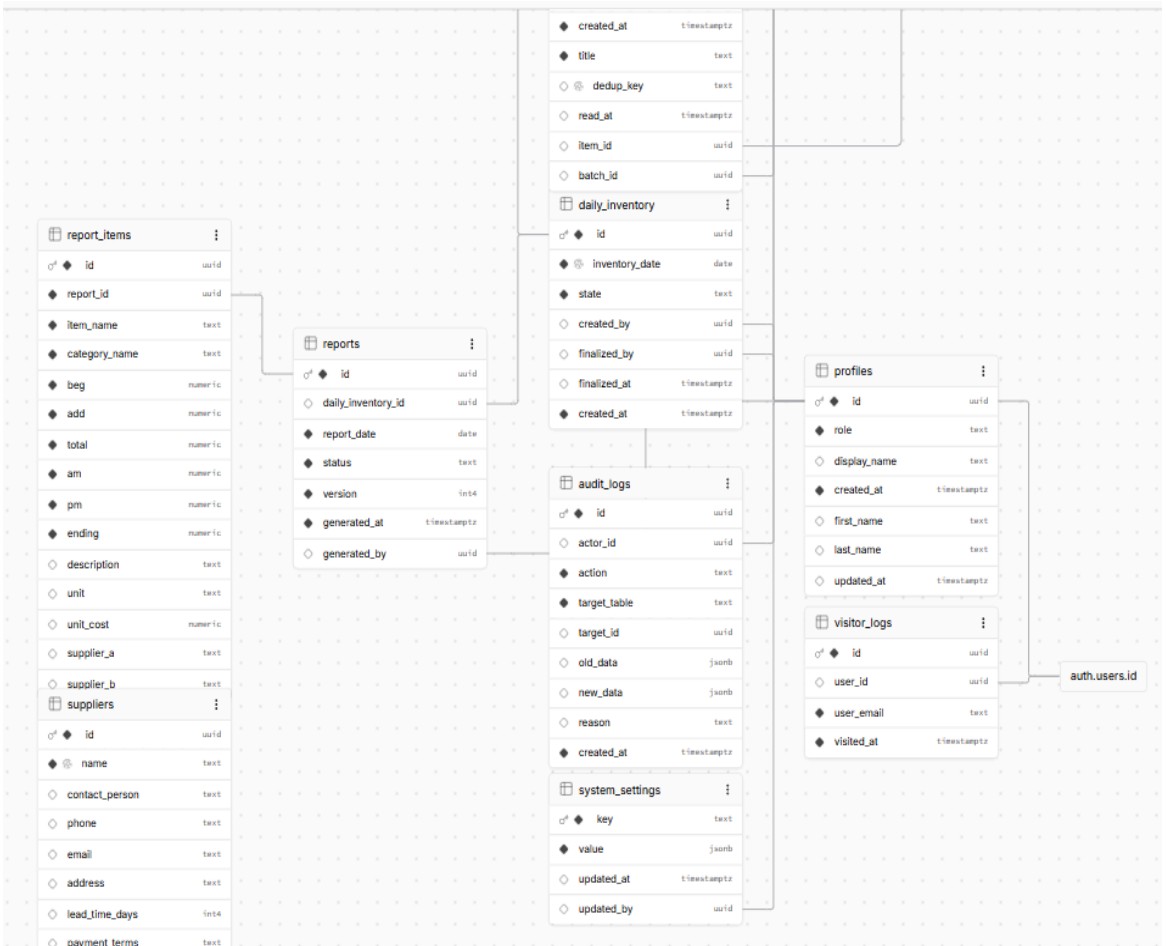
D. Stock Movement Input

- Source Station – The station releasing stock (*eg, Bodega*).
- Target Station – The station receiving stock (*eg, Kiosk or Grilled*).
- Quantity Transferred – Number of units moved between station locations.
- Movement Type – Action classification (*Stock-In, Station Transfer, Spillage, Waste*).

6.3 DATA SPECIFICATION

ENTITY RELATIONSHIP DIAGRAM





DATA DICTIONARY

TABLE NAME	TYPE	SCHEMA & DESCRIPTION
profiles	**TABLE**	id (UUID, FK auth.users), role (TEXT), display_name (TEXT), first_name (TEXT), last_name (TEXT), created_at (TIMESTAMPTZ), updated_at (TIMESTAMPTZ). Stores user roles and profiles.
categories	**TABLE**	id (UUID), name (TEXT), description (TEXT), created_at (TIMESTAMPTZ). Groups inventory items into categories.
inventory_items	**TABLE**	id (UUID), category_id (UUID, FK categories), name (TEXT), description (TEXT), unit (TEXT), unit_cost (NUMERIC), supplier_a (TEXT), supplier_b (TEXT), min_quantity (NUMERIC), is_active (BOOL), is_archived (BOOL), image_path (TEXT), created_at (TIMESTAMPTZ). Master catalog items.
stock_batches	**TABLE**	id (UUID), item_id (UUID, FK inventory_items), quantity (NUMERIC), expiry_date (DATE), received_date (DATE), created_at (TIMESTAMPTZ). Stores FEFO batch tracking info.

stock_movements	**TABLE**	id (UUID), item_id (UUID, FK inventory_items), batch_id (UUID, FK stock_batches), type (TEXT), quantity_before (NUMERIC), quantity_change (NUMERIC), quantity_after (NUMERIC), user_id (UUID, FK profiles), reason (TEXT), created_at (TIMESTAMPTZ). Audit trail for all inventory modifications.

daily_inventory	**TABLE**	id (UUID), inventory_date (DATE), state (TEXT), created_by (UUID), finalized_by (UUID), finalized_at (TIMESTAMPTZ), created_at (TIMESTAMPTZ). Tracks daily inventory sheets.
daily_inventory_items	**TABLE**	id (UUID), daily_inventory_id (UUID, FK daily_inventory), item_id (UUID, FK inventory_items), beg (NUMERIC), add (NUMERIC), total (NUMERIC), am (NUMERIC), pm (NUMERIC), ending (NUMERIC). Worksheet line items.
reports	**TABLE**	id (UUID), daily_inventory_id (UUID, FK daily_inventory), report_date (DATE), status (TEXT), version (INT4), generated_at (TIMESTAMPTZ), generated_by (UUID). Immutable report metadata snapshot.

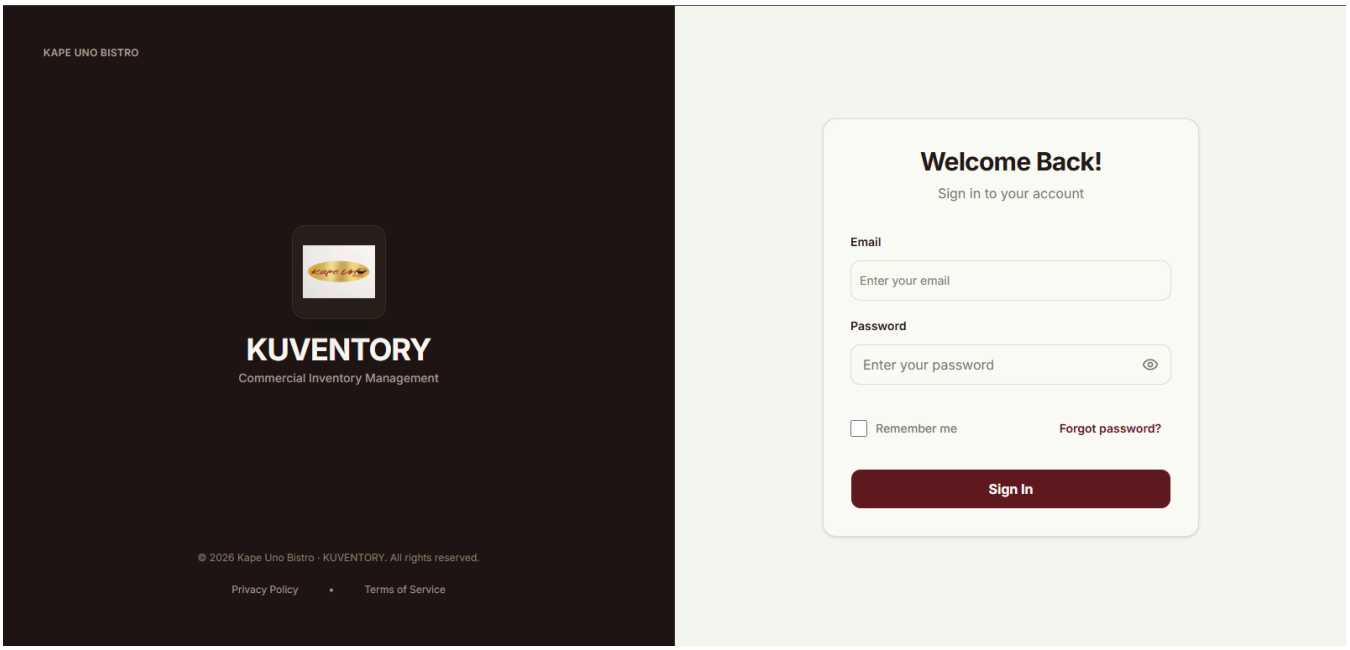
report_items	**TABLE**	id (UUID), report_id (UUID, FK reports), item_name (TEXT), category_name (TEXT), beg (NUMERIC), add (NUMERIC), total (NUMERIC), am (NUMERIC), pm (NUMERIC), ending (NUMERIC), description (TEXT), unit (TEXT), unit_cost (NUMERIC), supplier_a (TEXT), supplier_b (TEXT). Immutable report items snapshot.
notifications	**TABLE**	id (UUID), type (TEXT), target_id (UUID), message (TEXT), is_read (BOOL), user_id (UUID), created_at (TIMESTAMPTZ), title (TEXT), dedup_key (TEXT), read_at (TIMESTAMPTZ), item_id (UUID), batch_id (UUID). Low stock and expiration alerts.
audit_logs	**TABLE**	id (UUID), actor_id (UUID), action (TEXT), target_table (TEXT), target_id (UUID), old_data (JSONB), new_data (JSONB), reason (TEXT), created_at (TIMESTAMPTZ). Detailed security and data audit trail.
system_settings	**TABLE**	key (TEXT), value (JSONB), updated_at (TIMESTAMPTZ), updated_by (UUID). Global application configurations.

suppliers	**TABLE**	id (UUID), name (TEXT), contact_person (TEXT), phone (TEXT), email (TEXT), address (TEXT), lead_time_days (INT4), payment_terms (TEXT). Reference supplier directory.
visitor_logs	**TABLE**	id (UUID), user_id (UUID), user_email (TEXT), visited_at (TIMESTAMPTZ). Tracks system visits and sessions.

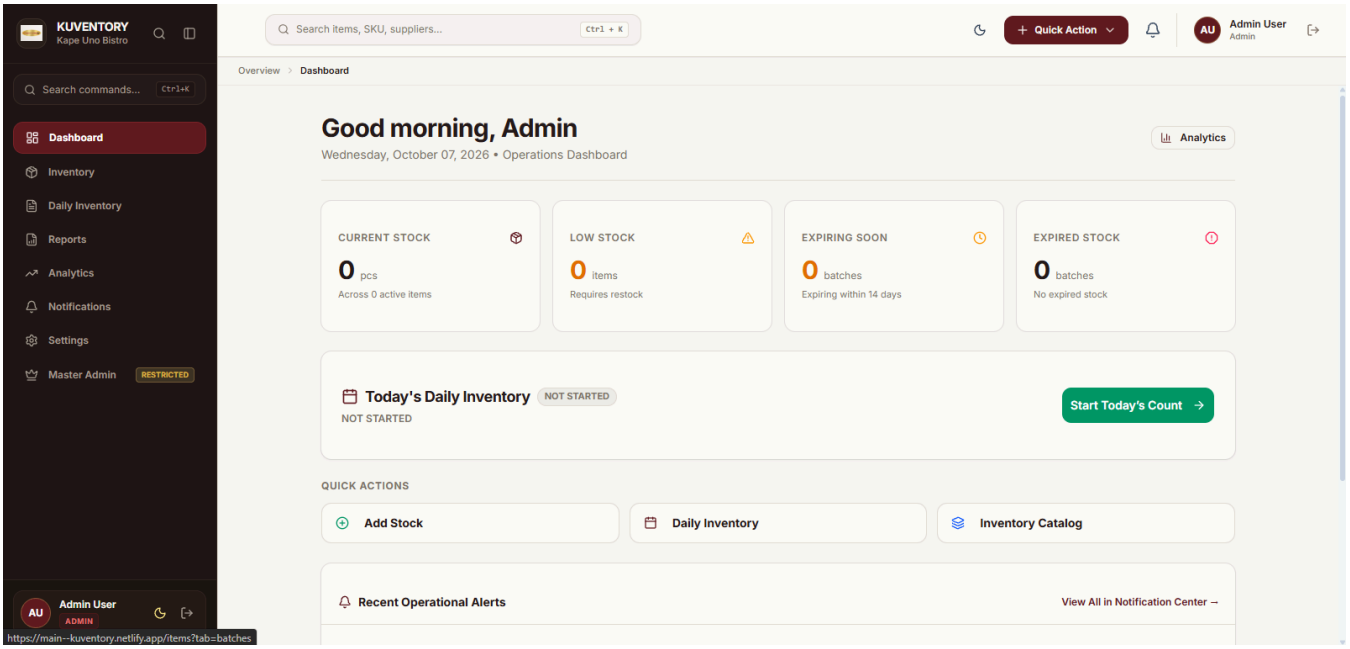
Figure 6.3.5.1 Table / File Layout

6.4 SCREEN LAYOUT SPECIFICATION

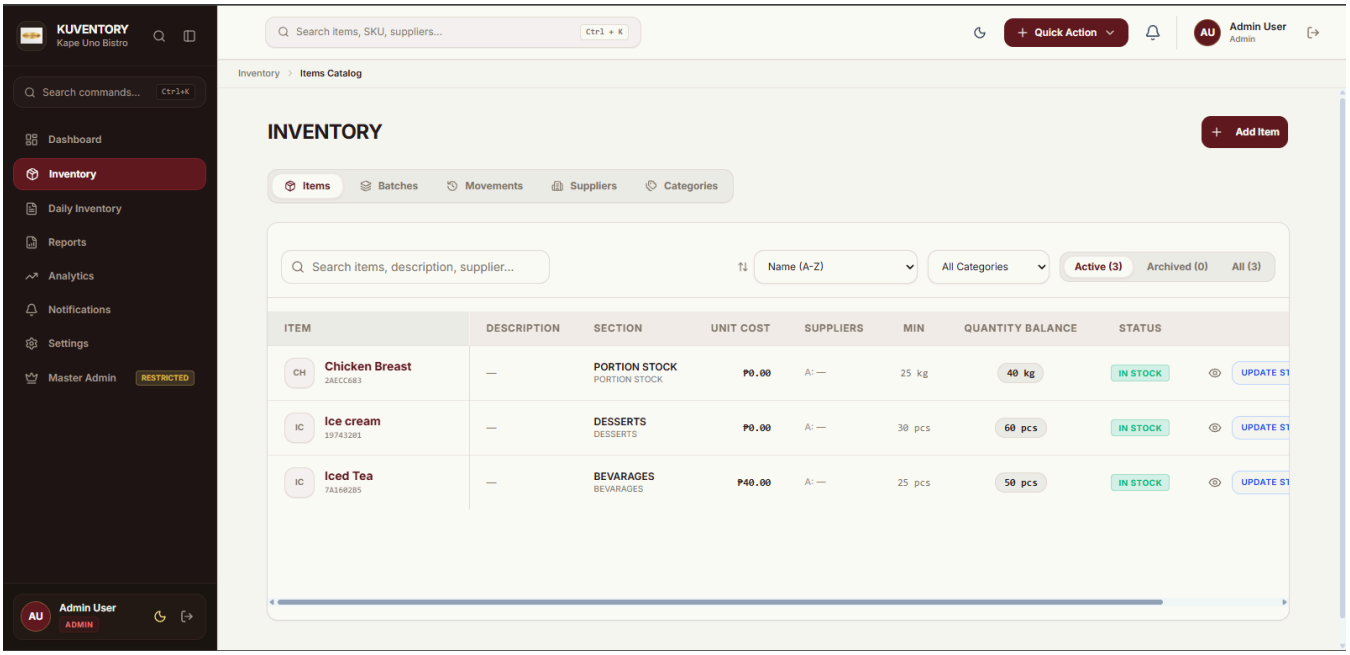
6.4.1 Screen layout specification



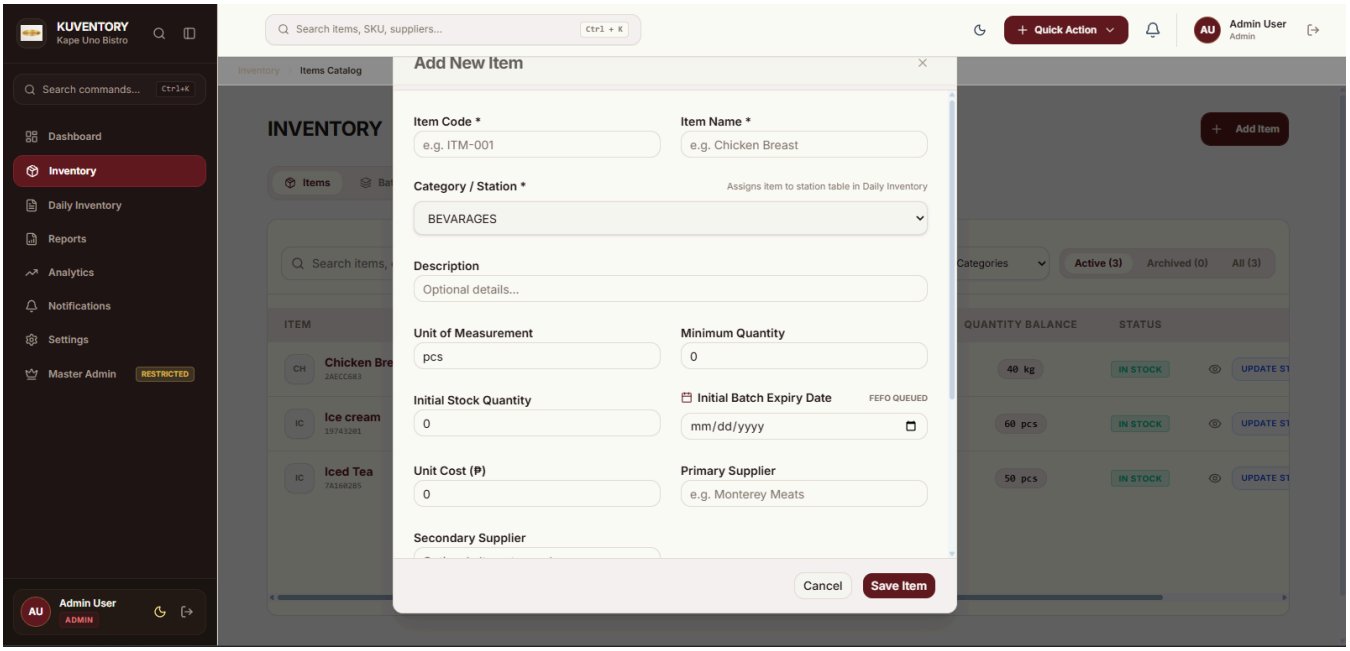
6.4.2 Dashboard



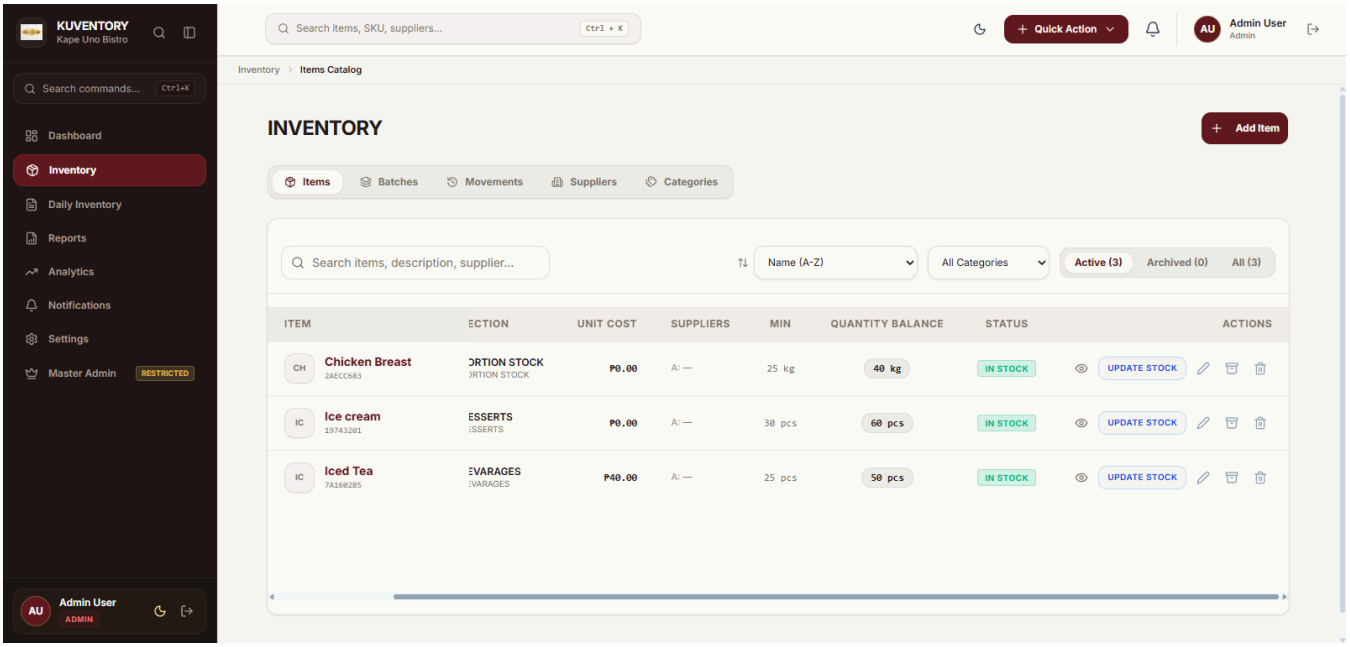
6.4.3 Inventory Management



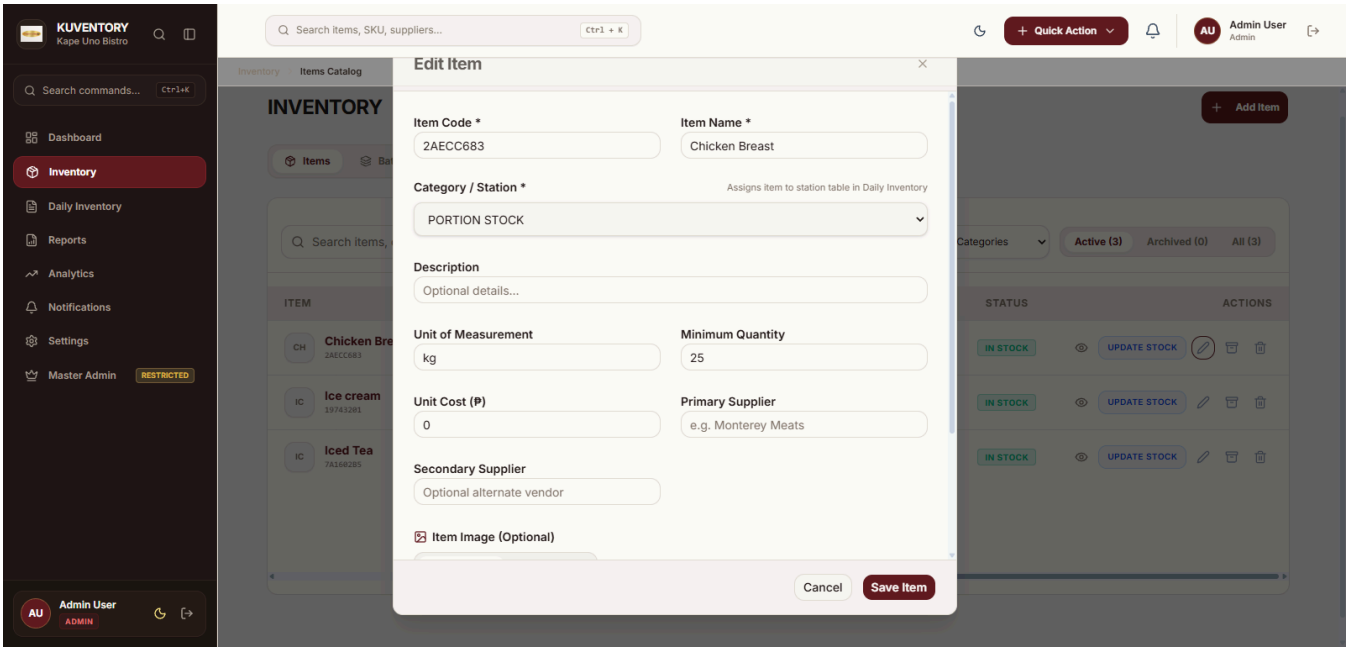
6.4.4 Add Inventory Item



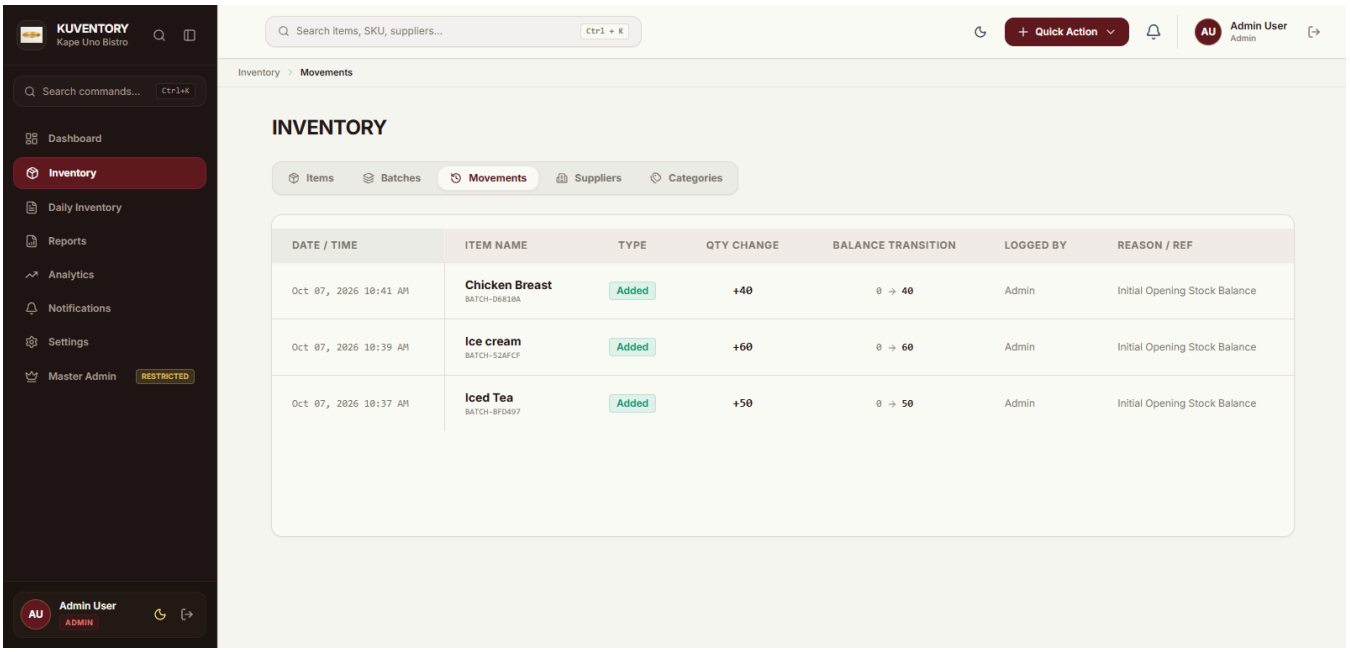
6.4.5 Edit / Delete / Archive Inventory



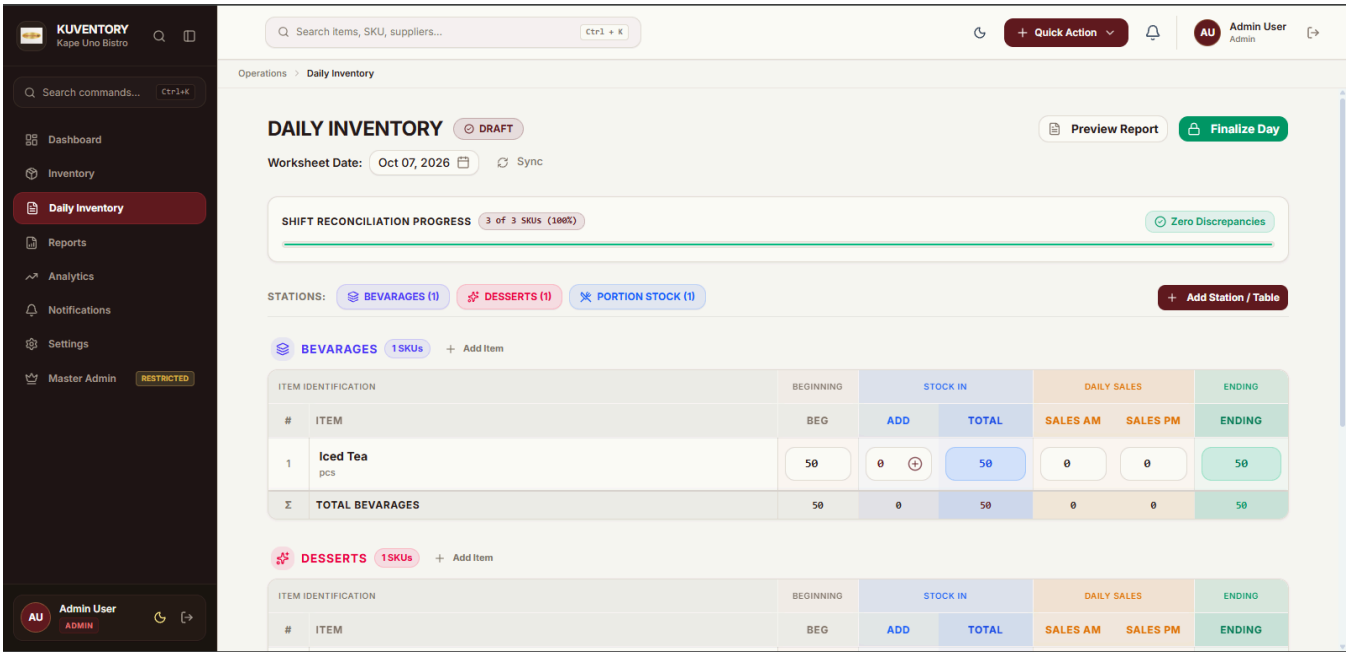
6.4.6 Begin Stock



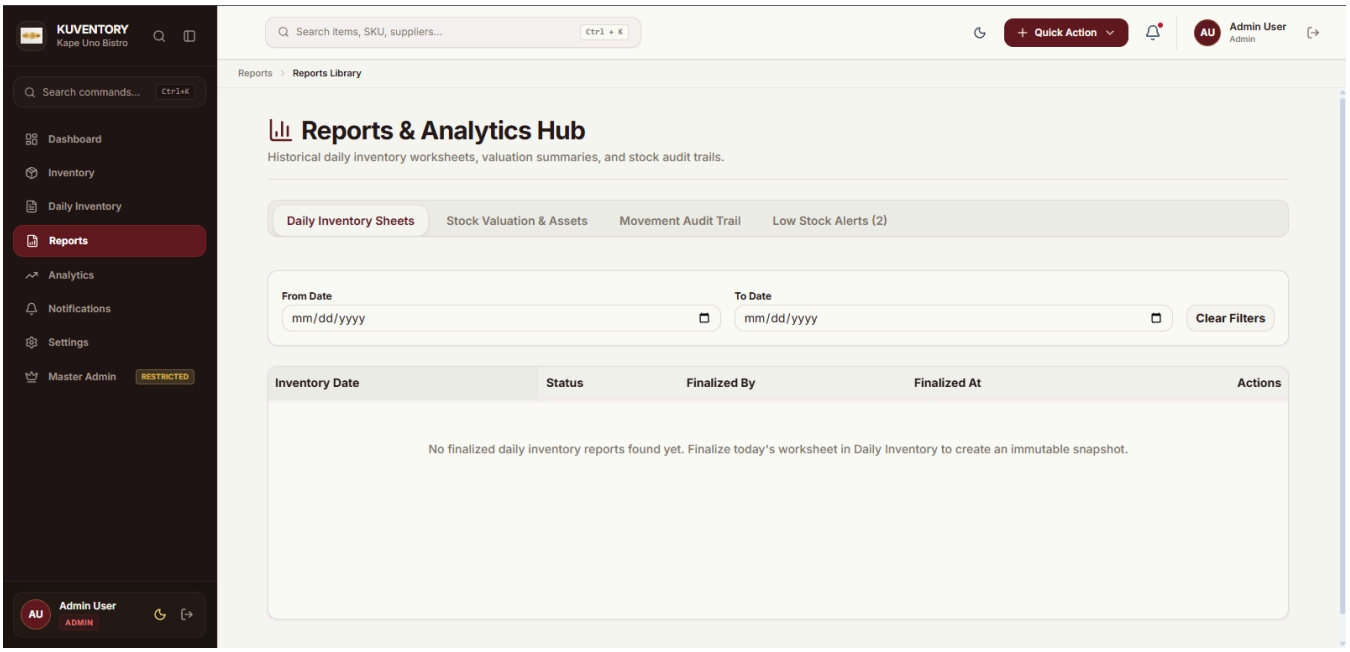
6.4.7 Stock in/Stock out



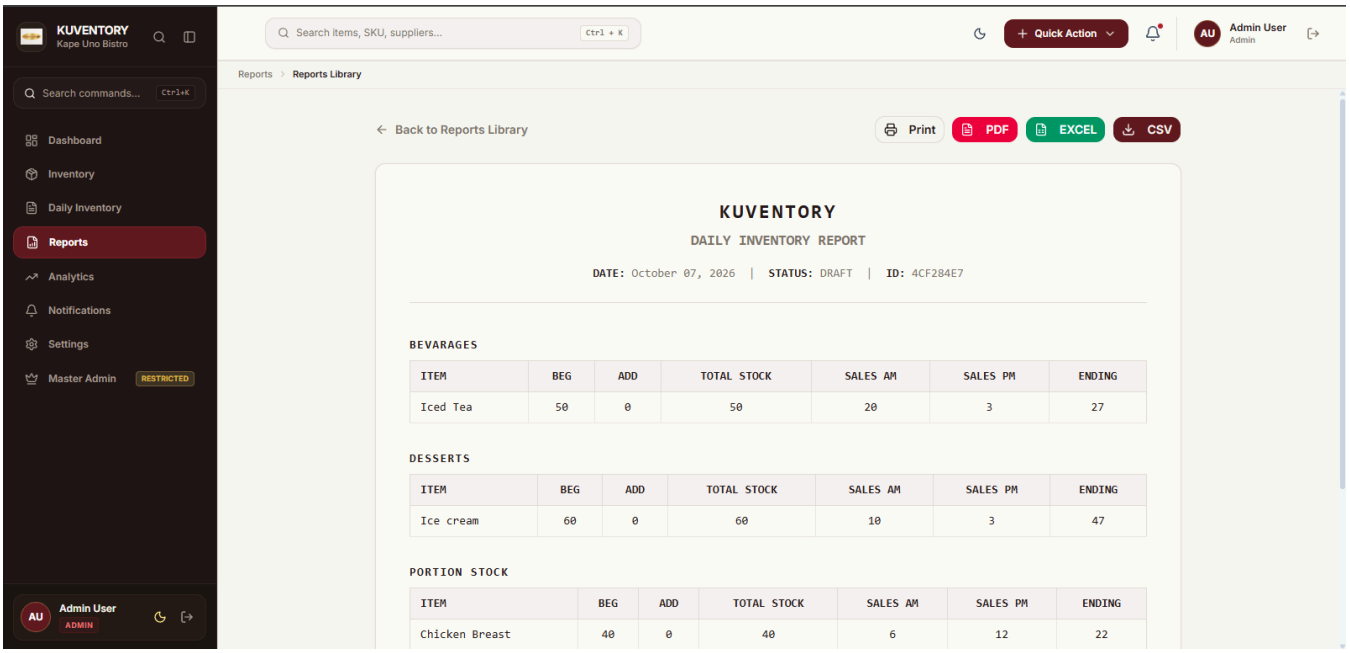
6.4.8 Daily Inventory Worksheet



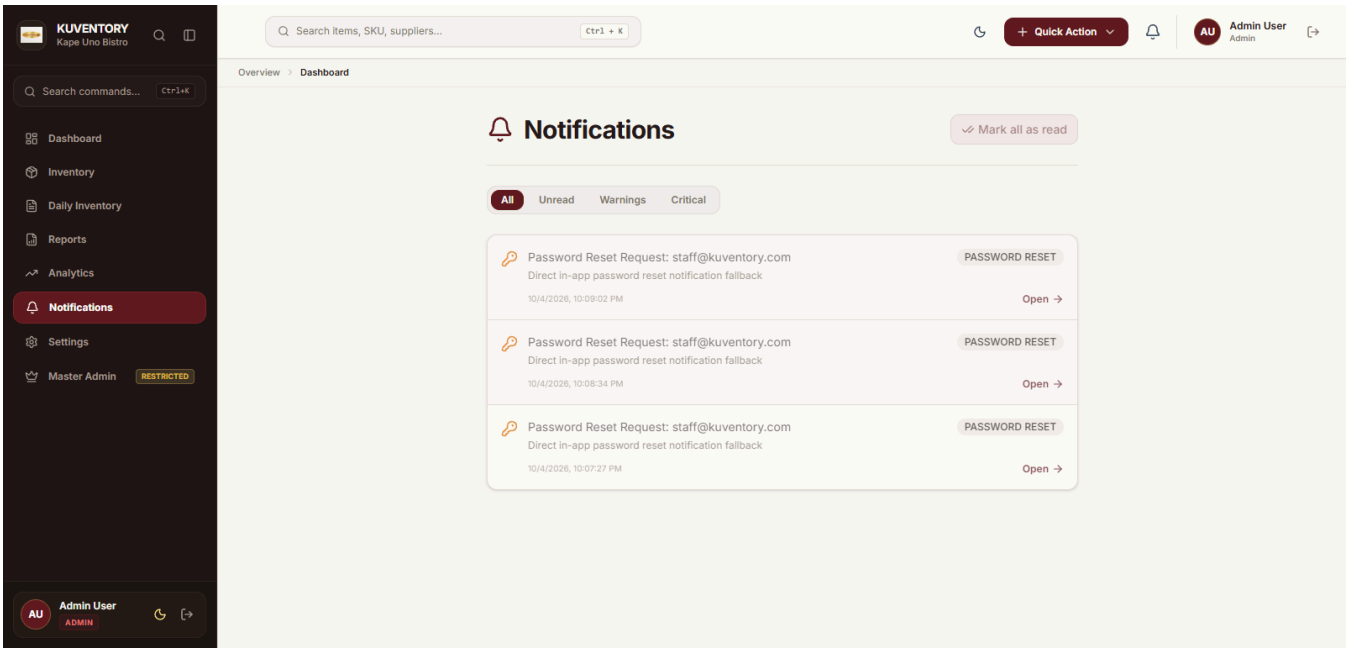
6.4.8 Reports



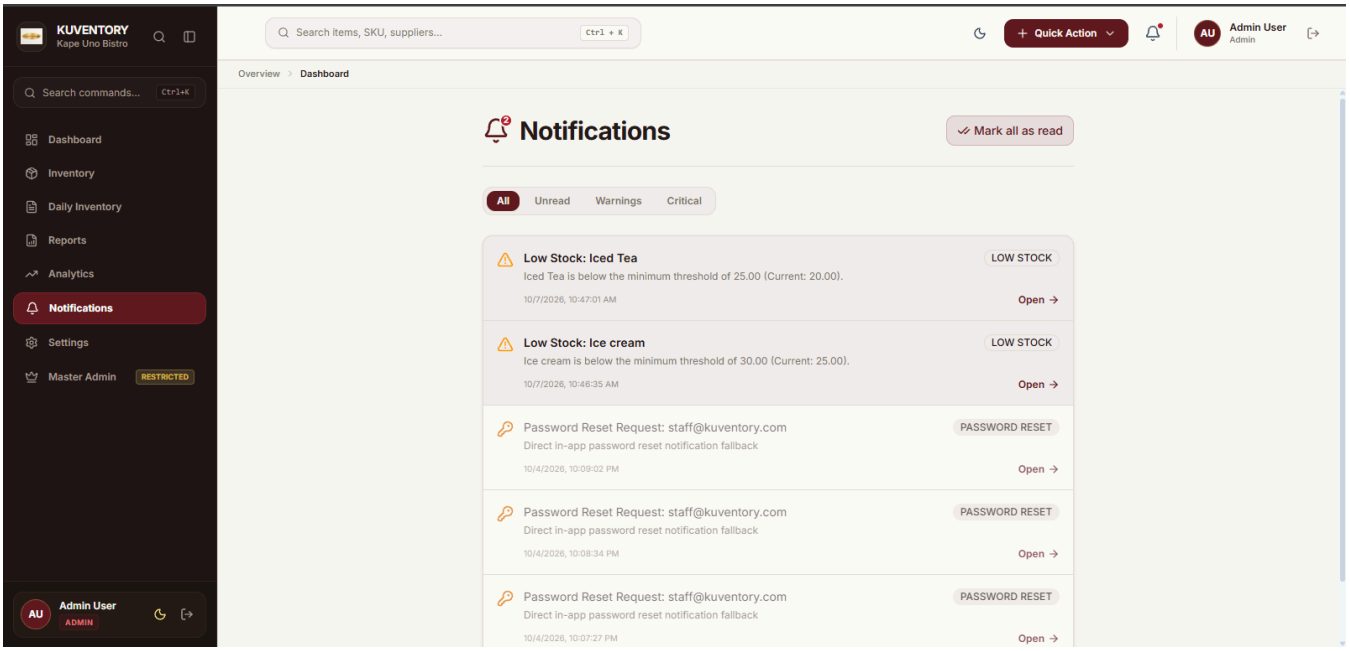
6.4.9 Export and Print Report



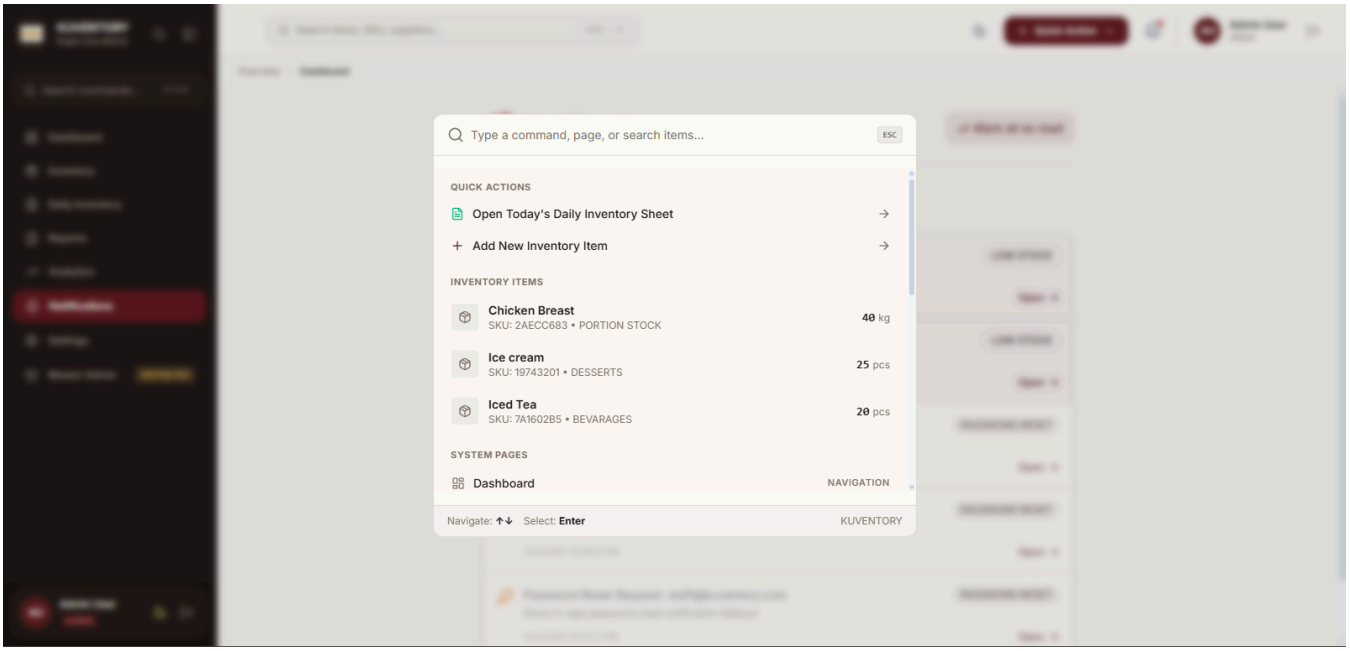
6.4.10 Notifications



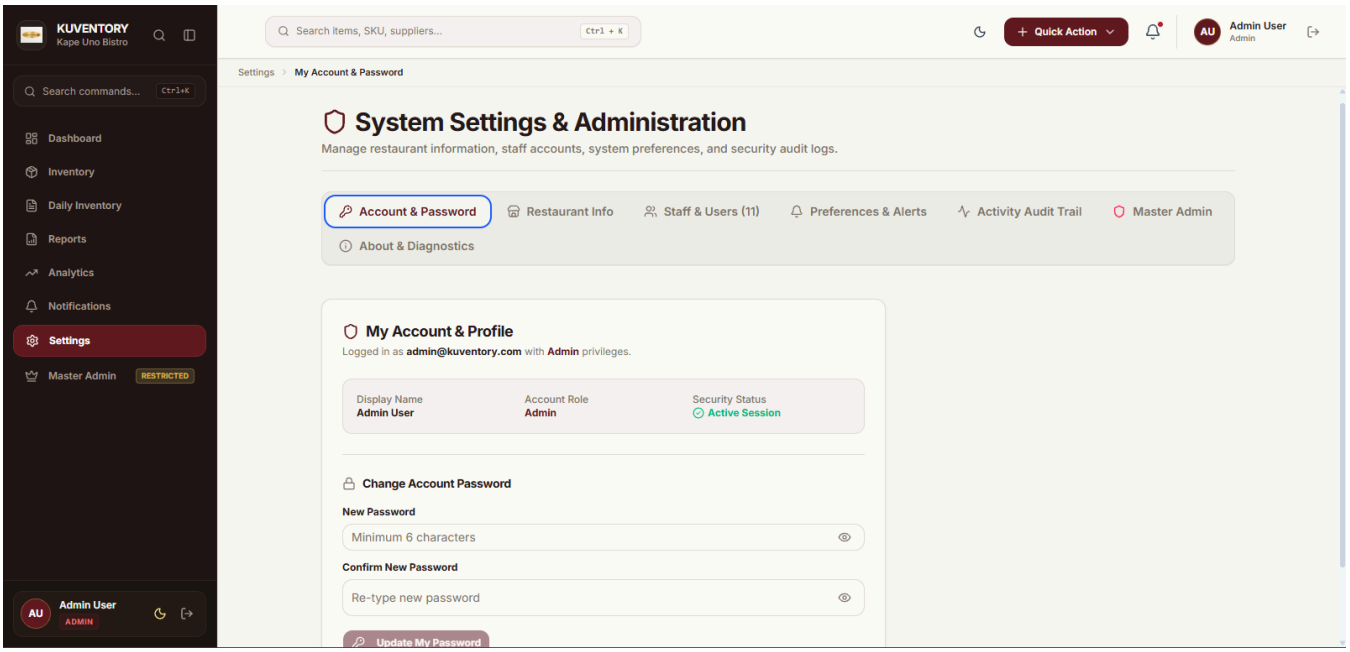
6.4.11 Low Stock Alert



6.4.12 Searching and Filtering



6.4.13 Account Management



6.4.14 Mobile Interface

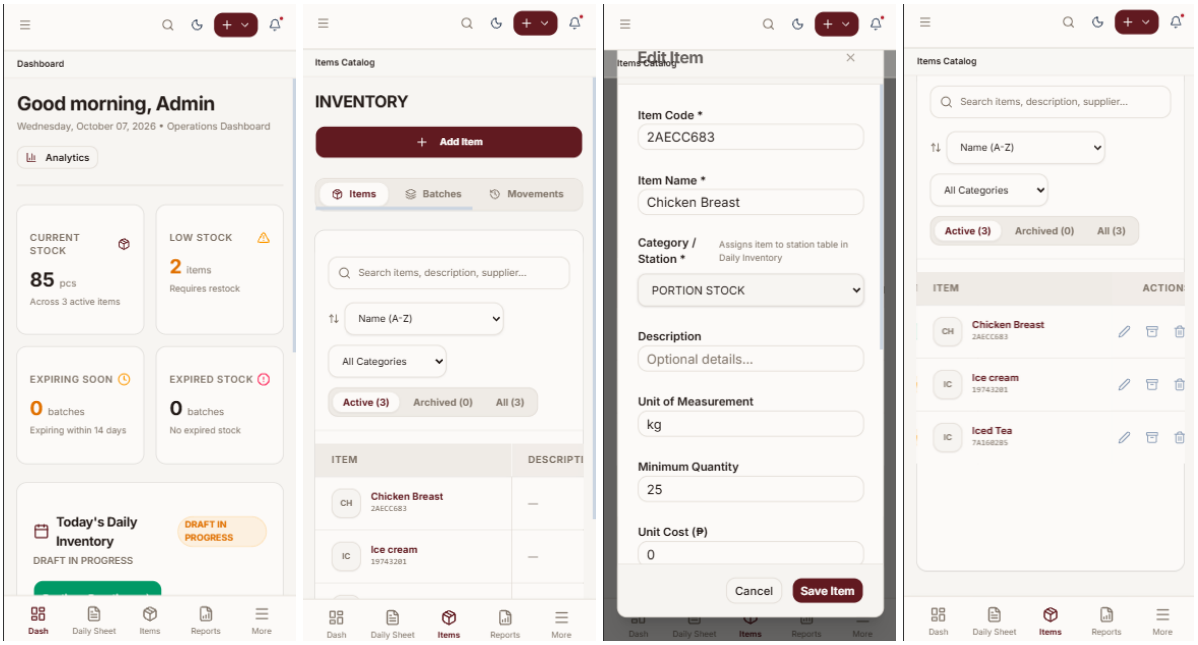


Figure: 6.4.14.1 Dashboard, Inventory, Add stock, and Edit/Archive/Delete

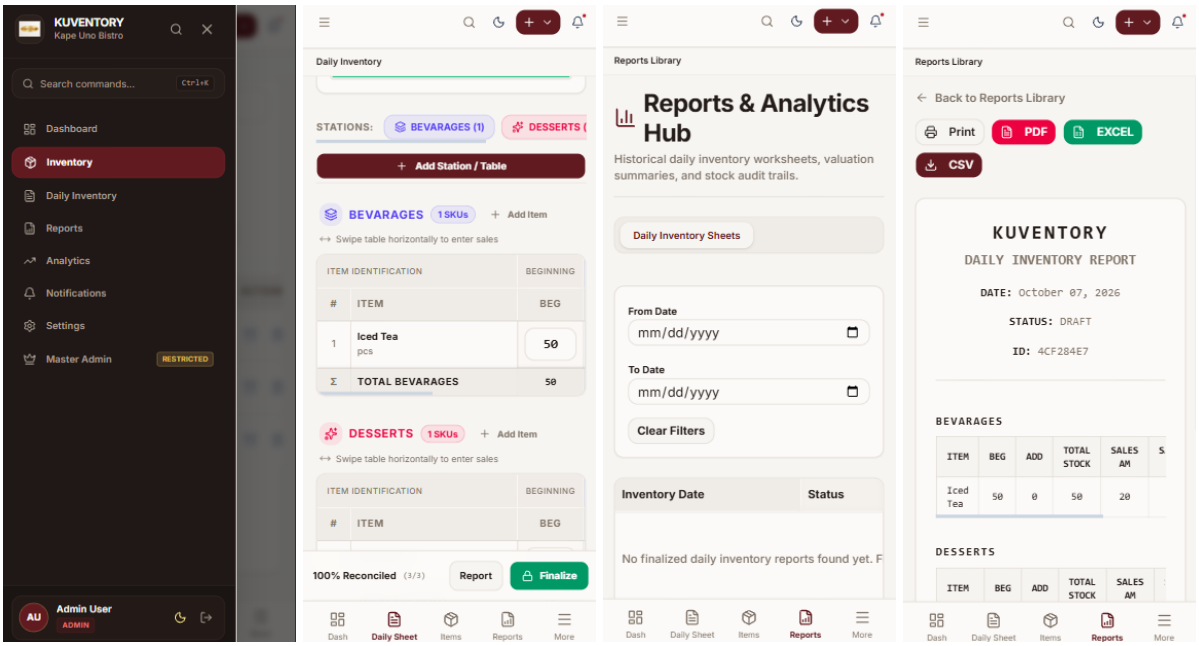


Figure: 6.4.14.2 Sidebar, Daily worksheet, Reports, and Print Reports

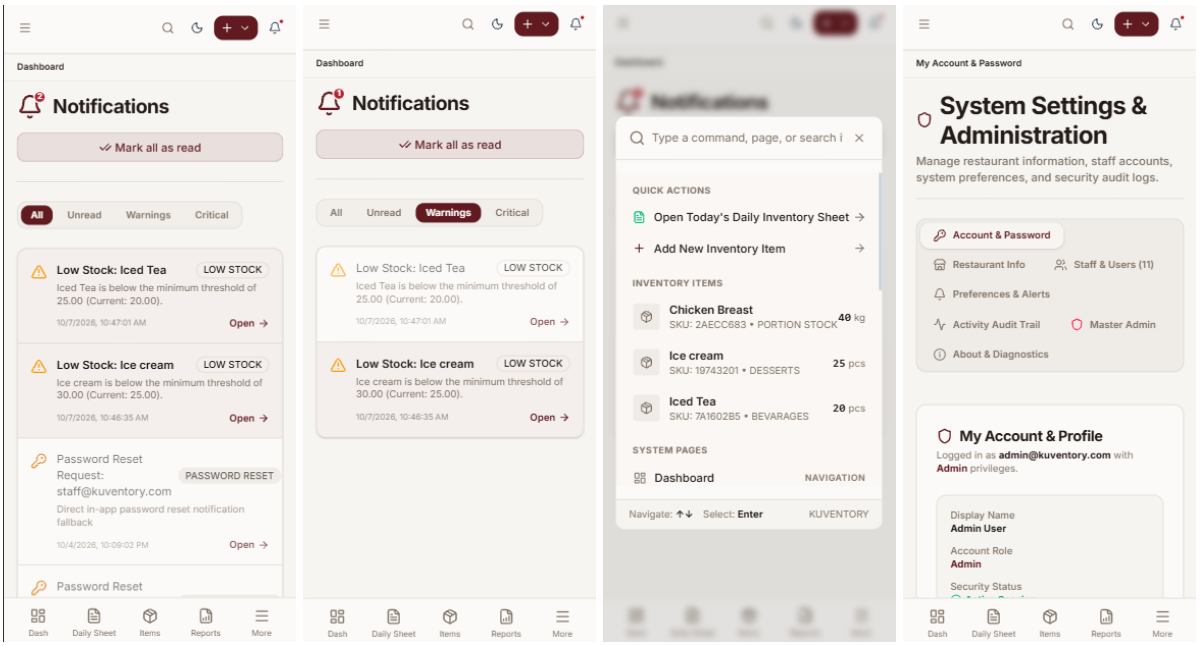
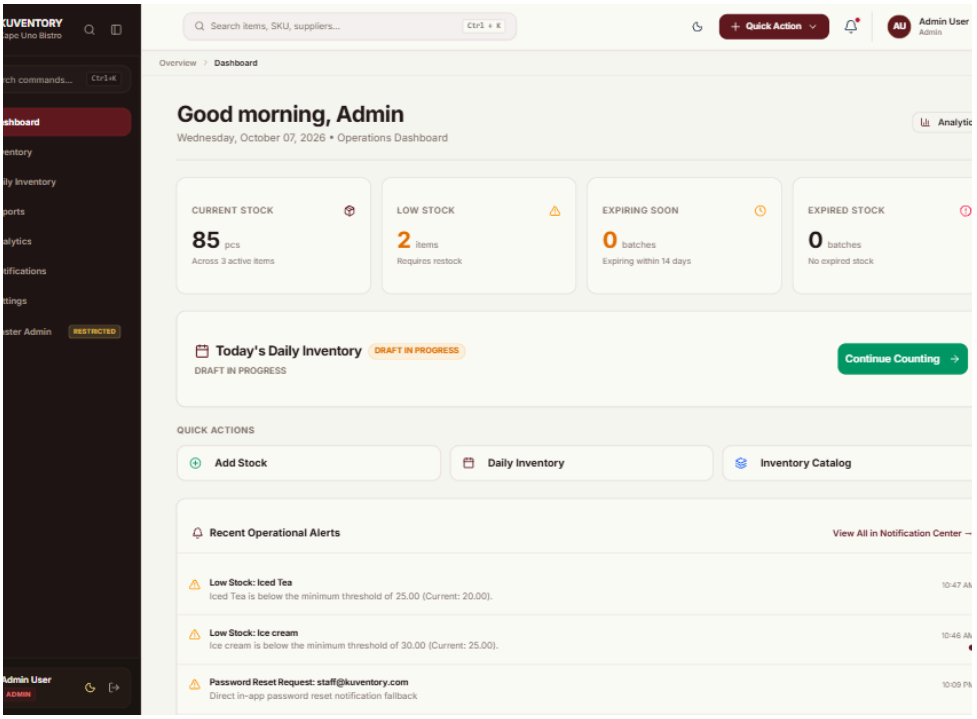


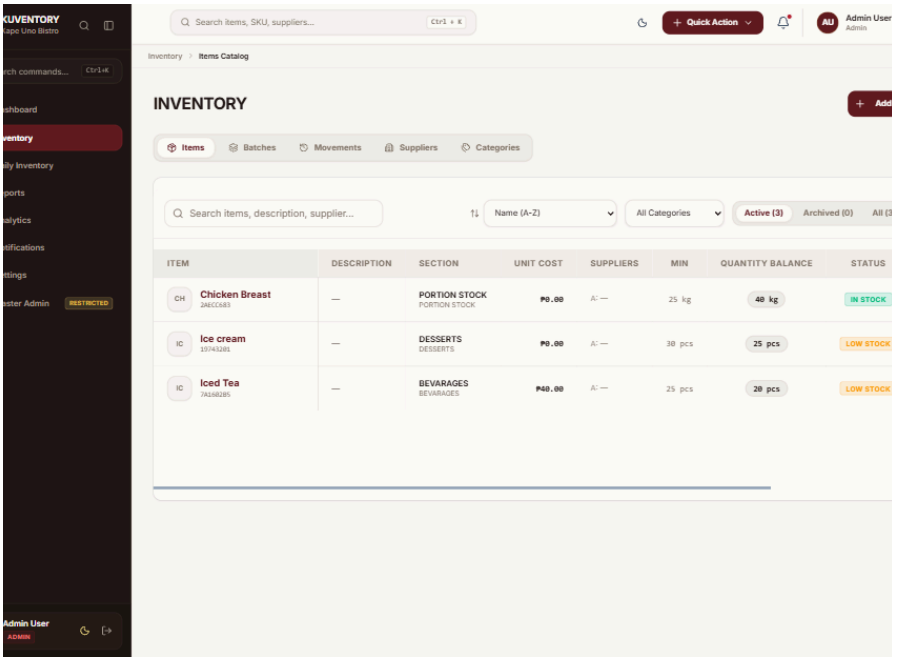
Figure: 6.4.14.3 Notifications, Low Stock reminder, Search Filtering, Account Management

6.4.15 Tablet/ Ipad Interface

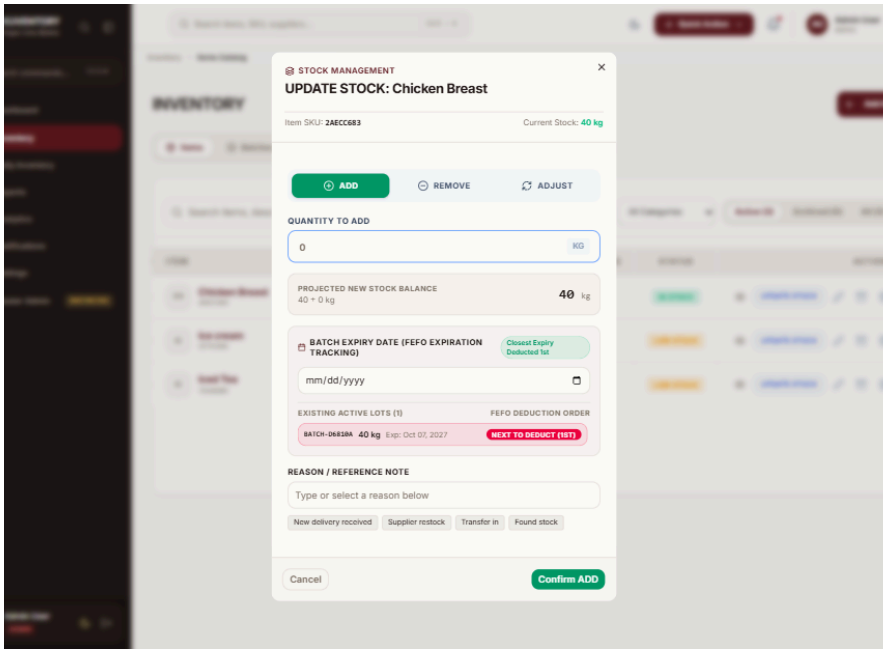
6.4.15.1 Dashboard



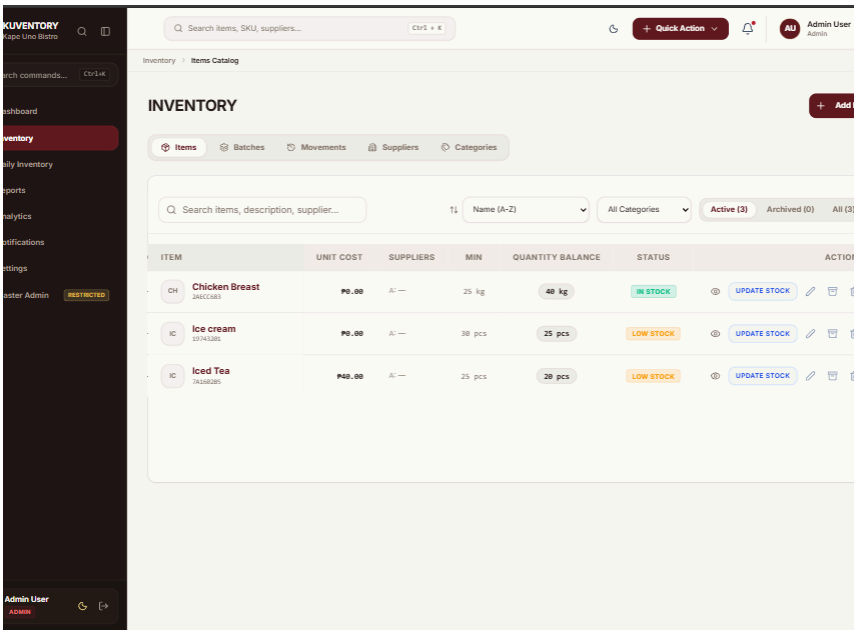
6.4.15.2 Inventory management



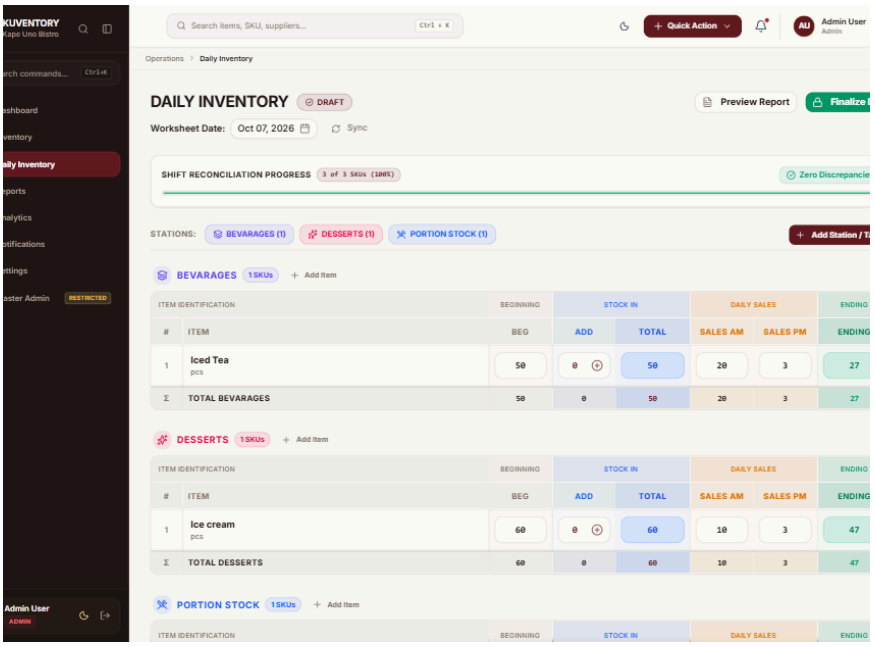
6.4.15.3 Add stock



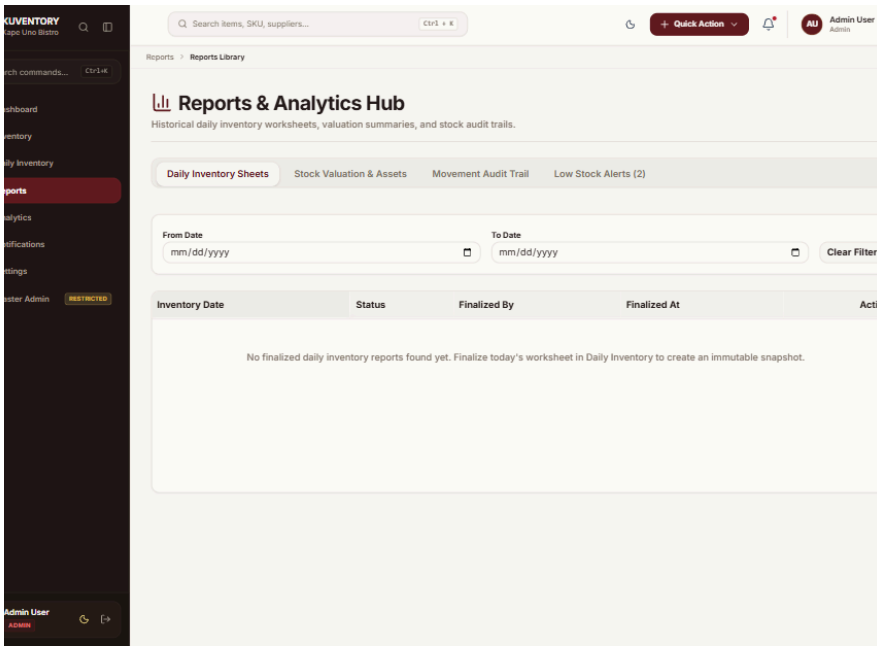
6.4.15.4 Edit/Archive/Delete



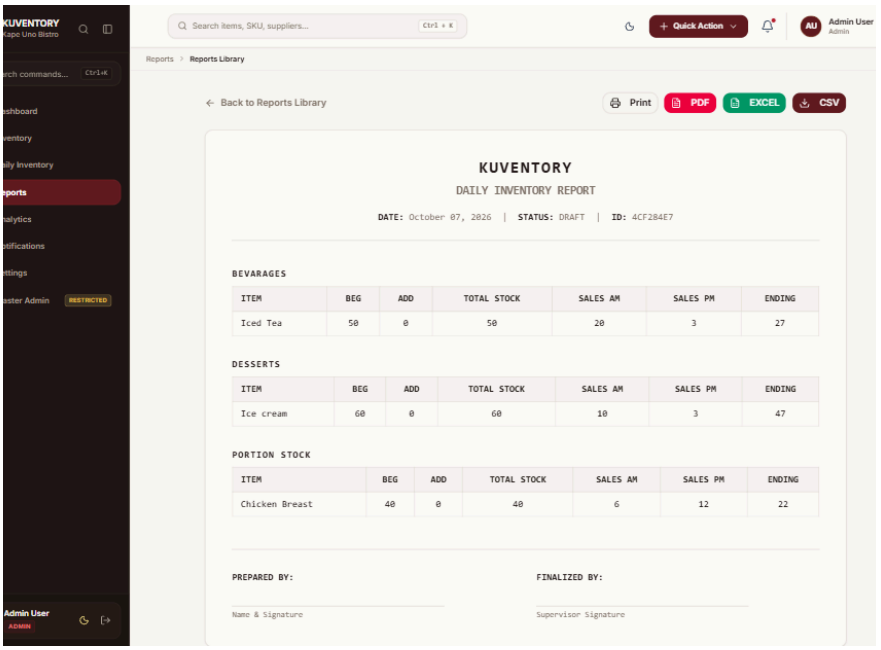
6.4.15.5 Daily worksheet



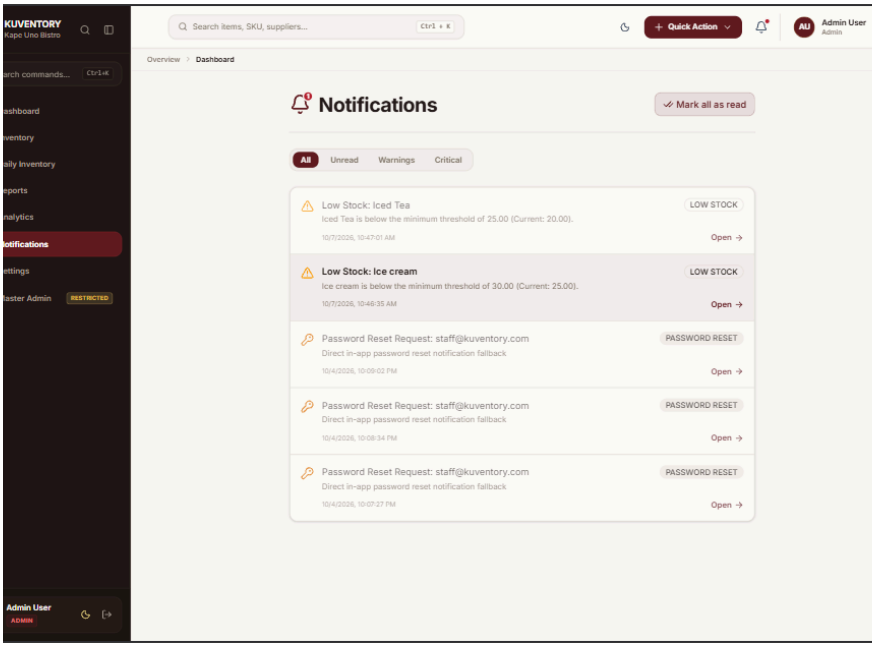
6.4.15.6 Reports



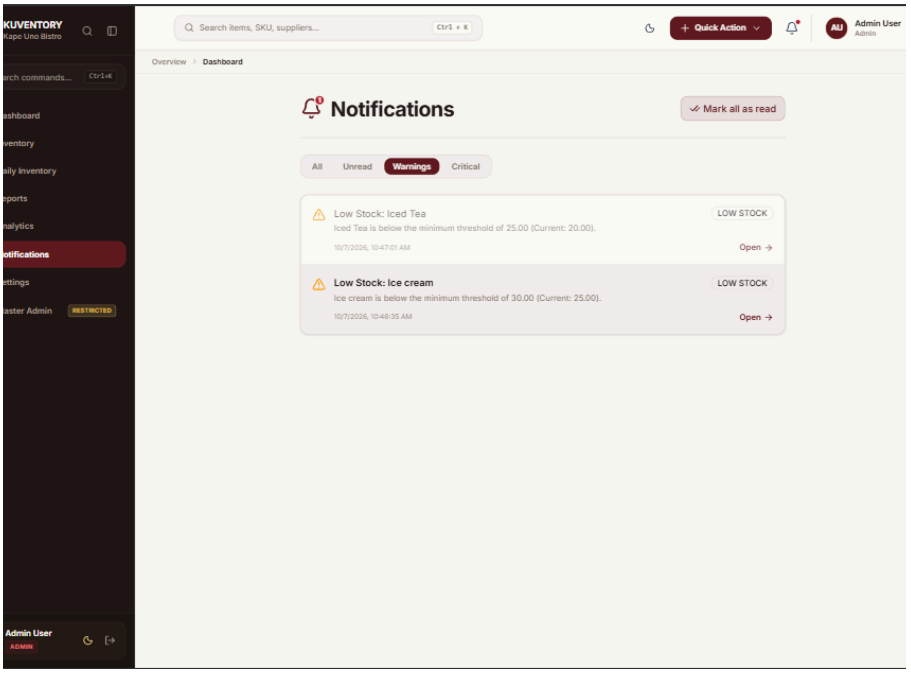
6.4.15.7 Report printing



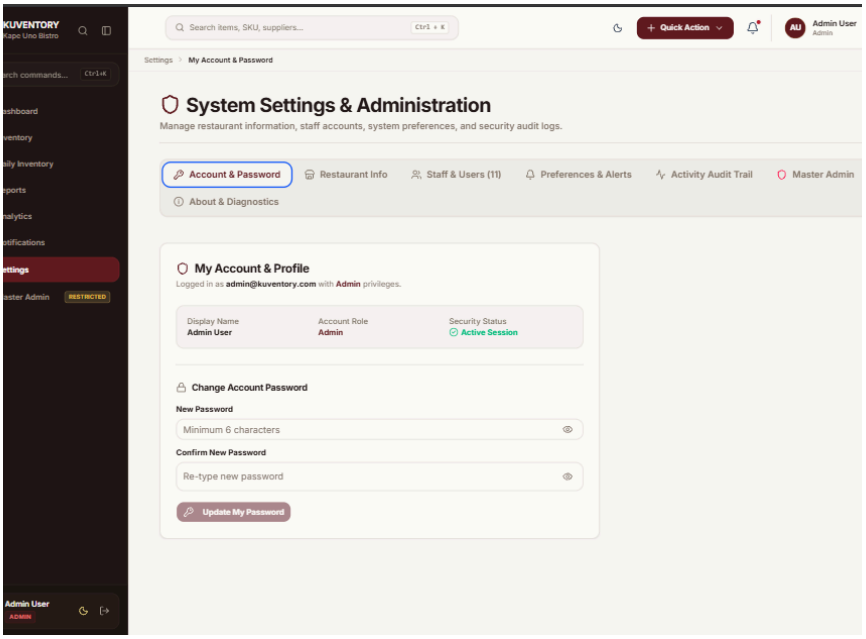
6.4.15.8 Notifications



6.4.15.9 Low stock alert



6.4.15.10 Account Management



6.5 REPORT / FORM SPECIFICATION

REPORT / FORM NAME	DESCRIPTION
Log in Form	Allows the owner to enter a registered username and password to securely access the KUVENTORY dashboard.
Product Form	Allows the owner to add, edit, and update product details such as category, SKU, cost price, selling price, and stock quantity.
Inventory Entry Form	Allows the owner to record beginning stock, added stock, sales, and ending stock for daily inventory monitoring.

Inventory Transfer Form	Allows the owner to transfer stocks between Bodega, Kiosk, Grilled, and Portion inventories.
Inventory Report	Generates a printable report showing current stock levels, inventory movements, low-stock items, and inventory valuation.
REPORT / FORM NAME	DESCRIPTION
Log in Form	Allows the owner to enter a registered username and password to securely access the KUVENTORY dashboard.
Product Form	Allows the owner to add, edit, and update product details such as category, SKU, cost price, selling price, and stock quantity.
Inventory Entry Form	Allows the owner to record beginning stock, added stock, sales, and ending stock for daily inventory monitoring.
Inventory Transfer Form	Allows the owner to transfer stocks between Bodega, Kiosk, Grilled, and Portion inventories.

7. SYSTEM CODING (PROTOTYPE)

KUVENTORY is a web-based inventory management system developed specifically to digitize and automate the operational inventory workflows of Kape Uno Restaurant. The system organizes major functions into stock monitoring, daily worksheet tracking, FEFO batch control, stock movements audit trails, immutable report generation, and system settings. The application is supported by a Supabase/PostgreSQL backend ensuring real-time synchronization and data persistence across modules.

The coding structure follows the existing KAPEUNO functionality. The frontend handles presentation, interaction, form input, and validation feedback; the application layer coordinates database mutations and queries; Supabase/PostgreSQL manages authentication, relational data, constraints, transaction integrity, and persistent records. The prototype is expected to maintain one authoritative source of truth for inventory and transaction data.

7.1 Prototype Functional Structure

Table 7.1 KAPEUNO prototype functional structure

MODULE	SYSTEM FUNCTION & FEATURES
User & Access Management	Supabase Auth login, session restoration, Role-Based Access Control (Admin vs Staff RLS policies), profiles management.
Dashboard Overview	Summary widgets for Total SKUs, Low Stock alerts, Out of Stock, Expiring Batches, Today's Worksheet status, and quick action shortcuts.
Master Items & Categories	CRUD operations for items, category mapping, station section assignment, unit costs, min thresholds, active/archived flags.
Daily Inventory Worksheet	Daily stock logging (Beg, Add, AM deduction, PM deduction, Ending Qty calculation), shift reconciliation, day locking upon finalization.

MODULE	SYSTEM FUNCTION & FEATURES
Batches & FEFO Engine	First-Expired, First-Out queue tracking, batch creation, expiration calculations, automated "USE FIRST" flagging.
Stock Movements Audit	Audit trail recording stock-in, stock-out, station transfers, spillage, and quantity_before / quantity_after values.
Immutable Reports Hub	Frozen report snapshots upon worksheet finalization, versioning, PDF/CSV export, historical valuation summaries.
Notifications Center	Automated alerts for low stock threshold breaches and impending batch expirations with deduplication key handling.
System Settings & Security	Global system configuration, user role assignments, audit logs tracking, password updates, visitor activity logging.

7.2 Coding and Data Flow

The prototype follows a controlled data flow: User Input → Frontend Validation → Application/Mutation Layer → Supabase → PostgreSQL → Database Result → Frontend Refresh → Related Module Updates. Persistent business data must not depend on temporary mock arrays or frontend-only state. When the database and frontend disagree, the persisted database state is authoritative and the frontend must refresh from it.

7.3 Inventory Prototype Relationship

The Inventory prototype is organized around the same item identity and authoritative stock information across the Daily Sheet, Master List, Low Stock, Sales, Dashboard, Reports, and History. Daily operating figures such as Beginning, Added, Morning deduction, Afternoon deduction, and Ending are connected to the item and movement records instead of creating an unrelated second inventory database.

7.4 System Testing and Debugging

System testing and debugging are part of the prototype stage. Each major function is inspected from the interface through the Supabase query or mutation and into PostgreSQL. Debugging focuses on root causes such as schema mismatch, invalid constraints, broken relationships, incorrect calculations, stale state, and incomplete cross-module updates.

Table 7.4.1 System testing and debugging process

DEBUGGING STAGE	ACTIVITY
Problem Identification	Document the affected KAPEUNO module, operation, record, database message, and observed result.
Data-Path Tracing	Follow frontend input → validation → service/mutation → Supabase → PostgreSQL → returned data → frontend state.
Schema and Logic Check	Compare frontend fields, types, constraints, foreign keys, relationships, RPCs, and calculations with the live database.
Root-Cause Fix	Correct the smallest necessary part of the implementation without removing the existing KAPEUNO feature.
Cross-Module Recheck	Recheck dependent modules such as Inventory, Sales, Dashboard, History, Reports, and Notifications.
Build and Release Check	Resolve build/configuration issues and prepare the corrected prototype for deployment and manual user testing.

7.5 Testing Stage

The testing stage verifies that the prototype behaves as an integrated system rather than as disconnected screens. Functional, integration, compatibility, usability, performance, security, database consistency, and recovery checks are included in the system plan. Final user acceptance testing is performed by the project team after the implementation and database repairs are complete.

8. SYSTEM TESTING PLAN

8.1 Testing Stage / Test Matrix

Table 8.2 Testing stage and debugging matrix

TESTING STAGE	EXPECTED OBSERVATION	TEST / CHECK	CORRECTIVE ACTION
Functional Testing	System functions complete the intended workflow and persist correctly.	Check valid/invalid inputs, CRUD actions, stock operations, transactions, and reports.	Fix failed functions and repeat the affected test.
Usability Testing	Users can find and operate KAPEUNO features with minimal confusion.	Observe navigation, form labels, tabs, search, messages, and common workflows.	Refine layout and messaging without changing feature purpose.
Compatibility Testing	The system operates on different devices and screen sizes.	Check Android, iOS, tablets, laptops, desktops, portrait, landscape, and responsive tables/charts.	Apply responsive corrections and recheck affected pages.
Integration Testing	Changes in one module appear correctly in related modules.	Trace Inventory ↔ Daily Sheet ↔ Sales ↔ Purchases ↔ Dashboard ↔ History.	Fix query, mutation, state, or relationship inconsistencies.

TESTING STAGE	EXPECTED OBSERVATION	TEST / CHECK	CORRECTIVE ACTION
Performance Testing	Queries and common tasks remain responsive.	Review query duration, repeated requests, payload size, and dashboard/report loading.	Optimize queries and duplicated application requests.
Security Testing	Authenticated access and database policies operate correctly.	Review session handling, RLS, protected routes, input validation, and client configuration.	Correct policy/configuration problems.
Database Consistency Testing	Database state and frontend state agree.	Check columns, constraints, foreign keys, inventory balance, totals, and history records.	Repair schema/mutation/relationship mismatches.
Disaster Recovery Testing	Backups and restore procedures are usable.	Perform controlled backup integrity and restore procedure checks.	Update recovery procedures and documentation.

8.2 Testing Schedule

Table 8.7.1 Testing schedule

ID	TASK NAME	DURATION	SCHEDULE	RESOURCE NAME
1	Functional Testing	14 Days	To be scheduled	KAPEUNO development and testing team

ID	TASK NAME	DURATION	SCHEDULE	RESOURCE NAME
2	Usability Testing	14 Days	To be scheduled	KAPEUNO development and testing team
3	Compatibility Testing	14 Days	To be scheduled	KAPEUNO development and testing team
4	Performance Testing	11 Days	To be scheduled	KAPEUNO development and testing team
5	Security Testing	12 Days	To be scheduled	KAPEUNO development and testing team
6	Integration Testing	10 Days	To be scheduled	KAPEUNO development and testing team
7	Database Consistency Testing	10 Days	To be scheduled	KAPEUNO development and testing team
8	Disaster Recovery Testing	7 Days	To be scheduled	KAPEUNO development and testing team

9 System Implementation Plan

9.1 Resource Requirements

9.1.1 Hardware Requirements

The hardware requirements are intended to support reliable browser access, database-backed transactions, report generation, and normal KAPEUNO operations. The system does not require a dedicated graphics processor for ordinary use.

Table 9.1.1 Hardware requirements

ENVIRONMENT	REQUIREMENT	PURPOSE
Developer / Hosting Side	Reliable cloud hosting and database services with sufficient CPU, memory, storage, and network capacity for expected users and transactions.	Supports application delivery and persistent database processing.
User Side	Desktop, laptop, tablet, or smartphone capable of running a modern web browser.	Supports daily KAPEUNO operations across devices.
Network	Stable internet connection.	Required for authenticated access, Supabase database operations, and Netlify deployment.

9.1.2 Software Requirements

Table 9.1.2 Software requirements

SOFTWARE / PLATFORM	SPECIFICATION	PURPOSE
Frontend Framework	React 19, TypeScript, Vite, Tailwind CSS, shadcn/ui, Lucide React, Recharts.	Responsive single-page application client.
Backend Infrastructure	Supabase Platform (PostgreSQL RDBMS, Supabase Auth, Row Level Security, RPC Functions, Triggers).	Database storage, security enforcement, and business logic execution.
Hosting & Deployment	Netlify for static client hosting; Supabase Cloud for database and backend services.	Production hosting and continuous delivery.

9.1.3 Human Resource Requirement

KAPEUNO is intended for authorized business staff who operate inventory and reporting functions. Administrative users maintain reference information and monitor system activity. The development/technical team maintains the database, deployment configuration, and system updates. Users require basic computer and browser skills supported by a user manual and demonstration session.

9.2.1 System Deployment

KAPEUNO is deployed as a web application using Netlify for the frontend and Supabase for authentication and persistent PostgreSQL-backed data. Deployment must use the same database contract, business rules, and configuration that were checked during development.

Table 9.2.1 System deployment activities

NO.	ACTIVITY	EXPECTED RESULT
1	Build the current KAPEUNO production frontend.	Latest source compiles and produces the production build.
2	Configure Netlify deployment and asset paths.	Deployed routes and assets load correctly.
3	Configure Supabase production settings.	Authentication and database requests use the correct project.
4	Verify schema, RLS, relationships, and required migrations.	Frontend queries and mutations match the live database.
5	Publish KAPEUNO v4.0.0.	The current version is deployed and visible in the application.

NO.	ACTIVITY	EXPECTED RESULT
6	Verify maintenance and recovery procedures.	Deployment can be maintained and recovered using documented procedures.

9.2.2 User's Manual and Personnel Training

Authorized KAPEUNO users shall receive a demonstration and guided practice before regular operation. The training session should explain the actual KAPEUNO workflows and emphasize how a change in one module can affect related records in Inventory, Sales, Purchases, Dashboard, History, Reports, and Notifications.

Table 9.2.2 User manual and personnel training

TRAINING AREA	ACTIVITY	EXPECTED COMPETENCY
Inventory	Create/edit items, Begin Stock, stock movements, Daily Sheet, Master List, Low Stock, History, Archived, and Restore.	Users can maintain correct inventory records.
Purchasing	Create purchases, enter quantities/costs, review totals, and process received stock.	Users can record purchases correctly.
Sales/Cashiering	Select items, verify stock, apply discounts, process payment, complete sales, and produce receipts.	Users can process sales consistently.
Expenses and Reports	Record expenses, review reports, search/filter data, and export results.	Users can monitor operating information and produce outputs.

TRAINING AREA	ACTIVITY	EXPECTED COMPETENCY
History, Notifications and Archive	Review activity, respond to low-stock alerts, view archived records, and restore records.	Users can monitor and recover information appropriately.
Authentication	Login, Logout, Remember Me, and protected system access.	Users can access KAPEUNO securely.

9.2.3 Implementation Schedule

Table 9.2.3 Implementation schedule

PHASES	ACTIVITY	DURATION / STAGE
Preparation and Planning	Verify device/browser requirements; verify network access; inspect project and Supabase configuration; review data backup and migration state; identify test data for controlled cleanup.	Planning stage
System Coding (Prototype)	Develop/refine KAPEUNO modules; connect frontend workflows to Supabase; implement inventory, purchasing, sales/cashiering, discounts, expenses, reports, history, notifications, and archive functions.	Coding / prototype stage
System Deployment and Configuration	Build production frontend; configure Netlify; configure Supabase production access; verify authentication, database schema, and deployment settings.	Deployment stage
Personnel Training	Conduct demonstration and guided practice; explain feature workflows; provide user manual.	Training stage

PHASES	ACTIVITY	DURATION / STAGE
Monitoring	Provide prompt technical assistance; monitor errors and user feedback; fix confirmed defects; improve performance and usability without removing intended KAPEUNO functions.	Maintenance stage

10. System Maintenance Plan

10.1 System Maintenance and Methodology

KAPEUNO maintenance follows a controlled cycle of identification, diagnosis, correction, verification, and release. A reported defect is first traced through the frontend, Supabase service layer, database schema, constraints, relationships, and dependent modules. Changes are made only after the intended KAPEUNO workflow is understood. Corrective work may include refactoring or reworking a broken module, but the functional purpose and tailored business rules are preserved.

Table 10.1 Maintenance methodology

MAINTENANCE TYPE	KAPEUNO APPLICATION
Corrective Maintenance	Fix confirmed errors such as failed saves, constraint violations, inconsistent stock, incorrect totals, broken notifications, failed exports, and history synchronization problems.
Adaptive Maintenance	Update responsive layouts, supported browsers/devices, environment settings, or deployment configuration when needed.
Perfective Maintenance	Improve performance, database query efficiency, state synchronization, report usability, and interface clarity without removing existing functions.

MAINTENANCE TYPE	KAPEUNO APPLICATION
Preventive Maintenance	Review migrations, constraints, dependencies, obsolete source files, logs, backup/recovery procedures, and recurring problem patterns.

10.2 System Testing and Debugging Plan

Table 10.2 System testing and debugging plan

TESTING STAGE	VERIFICATION FOCUS	EXPECTED RESULT
Functional Testing	Verify Create/Edit/Delete/Archive/Restore, Begin Stock, stock movements, purchases, sales, discounts, expenses, reports, export, notifications, history, and authentication.	All intended KAPEUNO functions operate and persist.
Integration Testing	Verify that information entered in one module is reflected in related modules, especially Inventory, Daily Sheet, Sales, Purchases, Dashboard, History, Reports, and Notifications.	Modules use consistent data and relationships.
Compatibility Testing	Verify the same feature set on desktop, laptop, tablet, Android, and iOS devices in portrait and landscape.	Layouts remain usable and no feature is lost.
Performance Testing	Verify common inventory queries, dashboard loading, searches, reports, and transactions under expected load.	Response remains acceptable during normal use.

TESTING STAGE	VERIFICATION FOCUS	EXPECTED RESULT
Security Testing	Verify authentication, protected access, RLS, input validation, session behavior, and safe client-side configuration.	Unauthorized access is prevented while valid users can operate the system.
Database Consistency Testing	Verify frontend values match database records, constraints are respected, stock has one authoritative source, and transaction totals are consistent.	Persistent records and related modules remain synchronized.
Disaster Recovery Testing	Verify availability of backup and restoration procedures for KAPEUNO data.	Data can be restored according to documented recovery steps.

10.3 Backup and Disaster Recovery

KAPEUNO shall maintain documented backup and restoration procedures for its persistent database information. Recovery activities should protect inventory, transaction, supplier, expense, history, notification, and configuration data. Backup integrity should be checked periodically and restoration should be performed in a controlled environment before being relied upon for an actual incident.

Table 10.3 Backup and Disaster Recovery Plan

RECOVERY ACTIVITY	DESCRIPTION	EXPECTED RESULT
Backup Procedure	Create and verify database backups according to the selected Supabase/database recovery process.	Backup completes and can be identified as valid.
Restore Procedure	Restore a known backup in a controlled environment and verify schema/data integrity.	Core KAPEUNO records can be recovered.
Recovery Check	Verify inventory, transactions, history, and related records after restoration.	Recovered records remain internally consistent.
Documentation	Keep recovery steps, responsibility assignments, and required configuration information updated.	The team can execute recovery without relying on undocumented knowledge.

10.4 System Maintenance Deliverables

- KAPEUNO v4.0.0 source code and production build.
- Supabase-connected schema, policies, functions, migrations, and application configuration required by the system.
- Updated user manual and personnel training materials.
- System testing and debugging records.
- Deployment and Netlify configuration.
- Backup and disaster recovery documentation.

- Maintenance and troubleshooting documentation.
- Updated system plan and testing schedule.

11.0 KEY PERSONNEL

11.1 PROJECT TEAM AND ORGANIZATIONAL STRUCTURE

The KU-VENTORY project team is a research and development team led by students and responsible for designing, developing, testing and implementation of "KU-VENTORY: Web-Based Inventory Management System with Real-Time Stock Alert for Kape Uno Bistro." The team uses their technical knowledge and theoretical principles to design and create a practical, efficient and user-friendly inventory management system.

The project aims to optimize the inventory operations of Kape Uno Bistro by developing an online platform that will help to control the stock level, monitor stock movement, batch number, expiration date using the First-Expired, First-Out (FEFO) technique, low-stock items and stock transfer between stations. Through collaboration of all project members, the system can increase efficiency, inventory precision and availability of information for authorized persons and management.

Nature of Reporting and Working Hours

The KUVENTORY project team follows a flexible work schedule based on the established project timeline, which includes specific phases, activities, responsibilities, and deadlines. Team members generally work from 8:00 AM to 5:00 PM, with additional working hours, when necessary, particularly during system development, integration, testing, documentation, and final project preparation. Each member is expected to provide regular progress updates and participate in scheduled meetings and project activities. Weekly reports and team consultations are conducted to monitor accomplishments, address challenges, coordinate assigned tasks, and ensure that project activities remain aligned with the established objectives and schedule.

Nature or Status of Employment

All members of the KUVENTORY project team are student developers participating in a temporary, project-based academic activity as part of their academic requirements. Their participation does not constitute formal employment, and the project operates under a non-salaried academic arrangement. Each member is assigned specific responsibilities according to their designated role and works collaboratively with the team to accomplish project objectives, meet deadlines, develop and test the system, prepare required documentation, and deliver the expected project outputs.

KEY PERSONNEL	TASK/RESPONSIBILITIES
Karla Ann DC. Calubaquib Project Manager	Oversees the overall KU-VENTORY project; prepares and monitors the project schedule; assigns and coordinates tasks among team members; facilitates meetings; monitors project progress and deliverables; coordinates with the client and stakeholders; manages project resources; identifies project risks and issues; and ensures that the project objectives are achieved within the agreed scope and timeline.

Mae Joy M. Acob Business / System Analyst	Gathers and analyzes the business and system requirements of Kape Uno Bistro; communicates with users and stakeholders to identify their needs; documents functional and non-functional requirements; analyzes existing inventory processes; prepares system specifications and process documentation; translates user requirements into clear system requirements; and coordinates with developers to ensure that the implemented system meets the identified business needs.
Johndel Christopher N. Aguinaldo System Developer	Designs, codes, and implements assigned system modules; develops system functionalities based on approved requirements and specifications; integrates frontend and backend components; performs debugging and error correction; maintains and improves source code; participates in database and system integration; and assists in system deployment and technical documentation.
Janine S. Marcos System Developer	Develops and implements assigned system modules and functionalities; writes, tests, and maintains source code; performs debugging and troubleshooting; assists in database and system integration; ensures that developed features comply with the approved system requirements; and participates in system testing, deployment, and technical documentation.

Michael James G. Riambon System Developer / UI/UX Designer	Develops and implements assigned system functionalities while designing the user interface and user experience of KU-VENTORY; creates interface layouts, navigation flows, and interaction designs; ensures consistency and usability across system pages; translates approved designs into functional interfaces; collaborates with other developers during integration; and assists in testing and improving the system's usability and functionality.
Meltritch Jan M. Zipagan Quality Assurance / System Tester	Develops and executes test cases based on system requirements; performs functional, integration, usability, and system testing; identifies, documents, and reports bugs and inconsistencies; verifies whether corrected issues have been properly resolved; monitors system quality and reliability; participates in user acceptance testing; and provides testing feedback to the development team for system improvement.

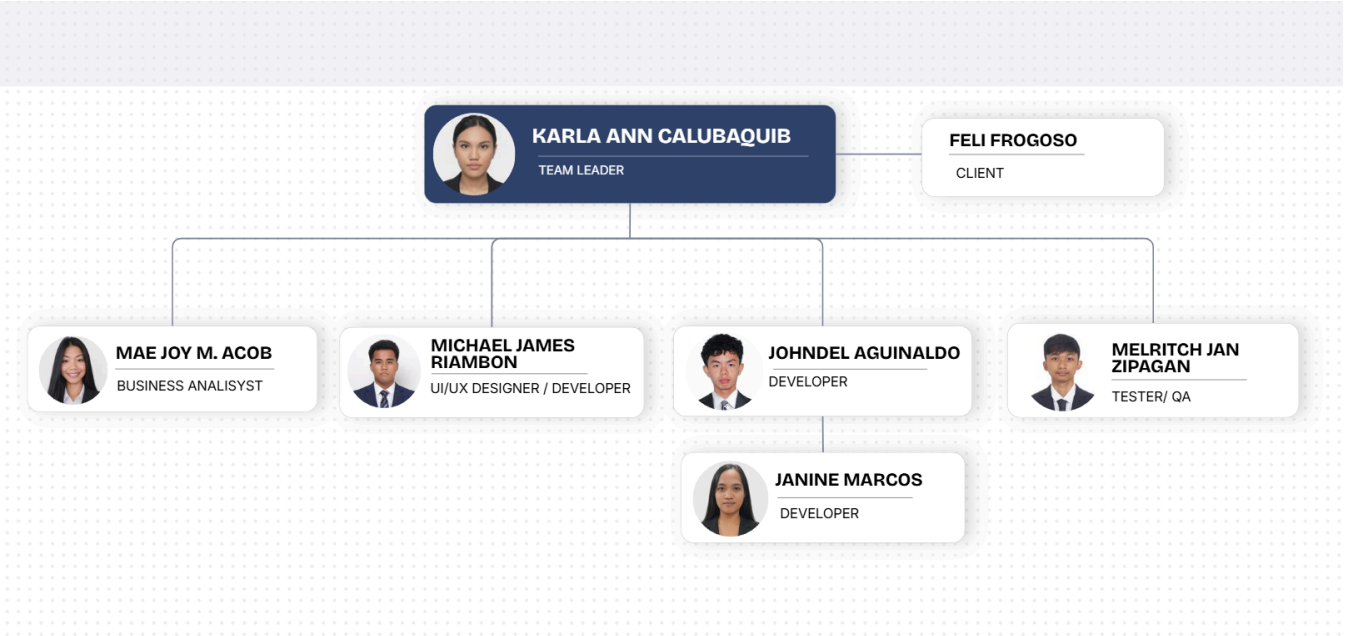


Figure 11.1 Organizational Chart

KARLA ANN DC CALUBAQUIB

Caggay, Tuguegarao City, Cagayan
Mobile Number: 09917101298
Email Address: calubaquibkarlaann@gmail.com



OBJECTIVES

Student with a background in technology, software development, and problem solving, pursuing a Bachelor of Science in Computer Engineering and a Team Leader. Dedicated to team leadership and project coordination where concepts and ideas are translated into practical solutions through effective teamwork. Capable communicator and collaborator who excels in organizing and solving technical problems.

PERSONAL INFORMATION

Age	20
Date of Birth	February 07, 2006
Birthplace	Tuguegarao City, Cagayan Valley
Sex	Female
Civil Status	Single
Religion	Christian Born Again

EDUCATIONAL BACKGROUND

Primary	Tuguegarao Northeast Cagayan Integrated School 2012 - 2018 Caggay/Tanza, Tuguegarao City, Cagayan Valley
Secondary	Cagayan National High School 2018 - 2024 Taft Street, Bagay Road, Tuguegarao City, Cagayan
Tertiary	Cagayan State University-Carig Campus Bachelor of Science in Computer Engineering 2024– Present Carig Sur, Tuguegarao City, Cagayan

SPECIAL SKILLS

Technical Leadership: Graphic Design, Stakeholder Management
Programming Languages: C++, Java, Microsoft Office Suite (Word, Excel, PowerPoint)
Development Tools: GitHub, Jira, Visual Studio

MICHAEL JAMES G. RIAMBON

Caritan Centro, Tuguegarao City, Cagayan
Mobile Number: 09525019551
Email Address: mikaeljamezz@gmail.com



OBJECTIVES

Innovative Full Stack Developer and UI/UX Designer with a strong foundation in Computer Engineering. Passionate about transforming complex product requirements and workflows into clean, accessible front-end layouts while ensuring scalable, efficient backend functionality through cloud technologies and relational databases.

PERSONAL INFORMATION

Age	20
Date of Birth	February 8,2006
Birthplace	Tuguegarao City, Tuguegarao
Sex	Male
Civil Status	Single
Religion	Roman Catholic

EDUCATIONAL BACKGROUND

Primary	Baggao East Central School 2012-2018 Zone 2, Hacienda Intal, Baggao, Cagayan
Secondary	Cagayan National High School 2018-2024 Taft Street, Tuguegarao City,Cagayan
Tertiary	Cagayan State University-Carig Campus Bachelor of Science in Computer Engineering 2024– Present Carig Sur, Tuguegarao City, Cagayan

SPECIAL SKILLS

Software Development & Databases: Python, C, C++, Java, Relational Databases (PostgreSQL)
Artificial Intelligence: Prompt Engineering & GenAI, Cloud AI (AWS/GCP)
Design & Research: Technical Research, Graphic Design, UI/UX Design
Professional Eligibility: Civil Service Professional Eligible

MAE JOY M. ACOB

Bubug, Santo Tomas, Isabela
Mobile Number: 09383240468
Email Address acobmaejoy@gmail.com



OBJECTIVES

Computer Engineering student aspiring to take on the role of a Business Analyst in a software development team. Focused on understanding business requirements, analyzing processes, and identifying opportunities for system improvement. Capable of gathering and documenting requirements, coordinating with stakeholders and developers, and helping ensure that software solutions meet business objectives and user needs.

PERSONAL INFORMATION

Age	19
Date of Birth	October 24, 2006
Birthplace	Santo Tomas, Isabela
Sex	Female
Civil Status	Single
Religion	Christianity

EDUCATIONAL BACKGROUND

Primary	Bubug Elementary School 2012-2018 Bubug, Santo Tomas, Isabela
Secondary	Calanigan National High School 2018-2024 Calanigan Norte, Santo Tomas, Isabela
Tertiary	Cagayan State University-Carig Campus Bachelor of Science in Computer Engineering 2024 – Present Carig Sur, Tuguegarao City, Cagayan

SPECIAL SKILLS

Productivity Tools: Microsoft Office (Word, Excel, PowerPoint)
Technical Skills: Requirements Gathering & Analysis, System Analysis
Professional Eligibility: Civil Service Professional Eligible

JOHNDEL CHRISTOPHER N. AGUINALDO

Nabannagan West, Lasam, Cagayan
Mobile Number: 09454518952
Email Address: ajohndel227@gmail.com



OBJECTIVES

Detail-oriented Computer Engineering student pursuing a Backend Developer role. Aiming to apply technical problem-solving, structured algorithm design, and software fundamentals toward building efficient backend logic, robust databases, and secure web services for high-performing systems.

PERSONAL INFORMATION

Age	20
Date of Birth	November 27, 2005
Birthplace	Lasam, Cagayan
Sex	Male
Civil Status	Single
Religion	Roman Catholic

EDUCATIONAL BACKGROUND

Primary	Nabannagan West Elementary School 2012-2018 Nabannagan West, Lasam, Cagayan
Secondary	Western Cagayan School of Arts and Trades 2018-2024 Centro 03, Lasam, Cagayan
Tertiary	Cagayan State University-Carig Campus Bachelor of Science in Computer Engineering 2024– Present Carig Sur, Tuguegarao City, Cagayan

SPECIAL SKILLS

Programming Languages: C++, Python, Java
Backend Development: RESTful API Development, SQL, Database Design
Version Control: Git, GitHub
Hardware Engineering: KiCad PCB Design and Fabrication
Development Tools: Visual Studio Code, PostgreSQL
Vocational Qualification: TESDA Driving NC II – Certified



JANINE S. MARCOS

Abra, Gattaran, Cagayan
Mobile Number: 09757700632
Email Address: marcosjanine3@gmail.com

OBJECTIVES

Computer Engineering student seeking a Backend Developer role. I want to use my programming, problem-solving, and database skills to build efficient and secure backend systems.

PERSONAL INFORMATION

Age	20
Date of Birth	June 27, 2006
Birthplace	Gattaran, Cagayan
Sex	Female
Civil Status	Single
Religion	Christianity

EDUCATIONAL BACKGROUND

Primary	Abra elementary school 2012-2018 Abra Gattaran ,Cagayan
Secondary	Gattaran National Trade School 2019-2024 Palagao Norte Gattaran, Cagayan
Tertiary	Cagayan State University-Carig Campus Bachelor of Science in Computer Engineering 2024– Present Carig Sur, Tuguegarao City, Cagayan

SPECIAL SKILLS

API Development – Building connections between applications
Database Management – Creating and managing databases
Problem-Solving - Finding and fixing coding errors

MELRITCH JAN M. ZIPAGAN

Caggay, Tuguegarao City, Cagayan
Mobile Number:
Email Address



OBJECTIVES

Detail-oriented Computer Science student with a strong foundation in software testing fundamentals, Java/Python programming, and automated testing tools built through academic coursework and personal projects. Adept at identifying edge cases, writing manual test scripts, and debugging web applications. Eager to apply problem-solving skills

PERSONAL INFORMATION

Age	20
Date of Birth	January, 20 2006
Birthplace	Tuguegarao, Cagayan
Sex	Male
Civil Status	Single
Religion	Roman Catholic

EDUCATIONAL BACKGROUND

Primary	Cabagan Science Elementary School 2011-2017 Catabayungan, Cabagan, Isabela
Secondary	Regional Science High School For Region II 2017-2023 Camp Samal, Arcon, Tumauini, Isabela
Tertiary	Cagayan State University-Carig Campus Bachelor of Science in Computer Engineering 2024– Present Carig Sur, Tuguegarao City, Cagayan

SPECIAL SKILLS

- **Exploratory Testing Mindset:** The ability to think like an adversarial user finding edge cases and break points that structured test scripts miss.
- **Empathy for the User Experience (UX):** Spotting visual inconsistencies, slow load times, or confusing UI flows beyond basic bug hunting.
- **Clear Technical Communication:** Translating bug impacts effectively to both non-technical project managers and backend developers.

12.0 Project Evaluation

Evaluation Approach

Project evaluation for KUVENTORY: Kape Uno Inventory Management System is divided into two continuous phases:

1. Throughout the Project (Formative Evaluation): Regular progress reviews, feature testing, and bug logging are conducted during system development to identify and address defects in inventory recording, stock movement processing, daily inventory reconciliation, FEFO batch tracking, and low-stock notifications. These activities help ensure that each module functions according to the system requirements.

2. At the End of the Project (Summative Evaluation): A formal software evaluation based on the ISO/IEC 25010 Software Quality Model is conducted with the system's intended users and relevant evaluators. The evaluation assesses KUVENTORY's functionality, performance, compatibility, usability, and overall suitability for managing inventory across Bodega, Kiosk, Portion, and Grilled.

PROJECT OBJECTIVES	EXPECTED RESULTS	KEY PERFORMANCE INDICATOR
User Authentication and Security	Secure login and role-based access control that restricts system features according to assigned user permissions.	• 100% of unauthorized access test cases are blocked. • Authorized users can access only the modules permitted by their roles.
Inventory and Item Management	Accurate and organized management of inventory items, categories, units of measurement, and stock records.	• All test items are saved and displayed correctly. • No duplicate or invalid records are accepted when prohibited by the system's validation rules.

Real-Time Stock Monitoring	Updated inventory quantities across Bodega, Kiosk, Portion, and Grilled following recorded stock transactions.	• 100% agreement between system balances and expected balances in controlled test scenarios. • Stock updates are reflected correctly after successful transactions.
Stock Movement Recording	Complete documentation of stock additions, deductions, adjustments, and transfers, with traceable movement history.	• 100% of successfully recorded test transactions appear in the movement history. • Recorded quantities, locations, dates, and transaction details match the submitted transactions.
FEFO Batch and Expiration Tracking	Organized batch monitoring that prioritizes items with the earliest expiration dates for use or release.	• Items with earlier expiration dates are correctly prioritized in test scenarios. • Batch expiration information is displayed accurately.
Daily Inventory Reconciliation	Digital recording and reconciliation of daily inventory quantities to help identify stock discrepancies.	• Daily inventory records are saved and retrieved correctly. • Calculated stock differences match the expected results in test cases.
Inventory Alerts and Reports	Timely identification of low-stock, out-of-stock, and near-expiration items, with accurate inventory reports.	• Alerts are triggered according to configured stock thresholds and expiration rules. • Generated reports match the corresponding inventory records.
Interface Usability and Compatibility	A user-friendly, responsive web interface accessible through supported desktop and mobile browsers.	• Overall user evaluation rating of at least 4.0 out of 5.0. • Core inventory functions operate correctly on supported browsers and screen sizes.

Project Objectives, Results, and Key Indicators The following table presents

KUVENTORY's main project objectives, expected results, and measurable performance indicators.**Table 12.1 Project Objectives, Expected Results, and Key Performance Indicators**

Monitoring and Evaluation (M&E) Schedule

The M&E activities are conducted according to the following timeline:

PHASE	**M&E ACTIVITY**	**FREQUENCY**	**METHODS & TOOLS**	**TARGET OUTPUT**
Development	Sprint & Milestone Tracking	Weekly	Team progress meetings, task monitoring, and GitHub commit	Verified weekly deliverables and updated development progress
Integration	Internal Module Testing	Daily	Test cases, test scripts, Supabase database checks, and bug logs	Tested inventory modules and documented, resolved defects
Pre-Deployment	User Acceptance Testing (UAT)	End of Development	Scenario-based walkthroughs of stock recording, stock transfers, FEFO tracking, and inventory reconciliation	Completed UAT records and documented user feedback

Post-Deployment	Summative Evaluation Survey	Final Week	ISO/IEC 25010-based evaluation questionnaire using a 5-point Likert scale	Evaluation results, computed mean scores, and software quality assessment

Table 12.2 Project Monitoring and Evaluation Schedule

Evaluation Instrument and Scoring Scale

The evaluation used an **ISO 25010 Software Quality Evaluation Questionnaire** containing 30 indicators covering the 8 software quality dimensions. Responses were scored using the following 5-point Likert scale:**Table 12.3 Five-Point Scale Scoring and Interpretation Guide**

SCALE	VERBAL INTERPRETATION
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

Summary of Evaluation Results (ISO 25010)

ISO 25010 SOFTWARE QUALITY STANDARD	OVERALL MEAN	VERBAL INTERPRETATION
1. Functional	4.55	Strongly Agree

Sustainability		
2. Reliability	4.45	Strongly Agree
3. Portability	4.55	Strongly Agree
4. Usability	4.5	Strongly Agree
5. Performance Efficiency	4.6	Strongly Agree
6. Security	4.5	Strongly Agree
7. Compatibility	4.5	Strongly Agree
8. Maintainability	4.65	Strongly Agree
OVERALL COMPOSITE MEAN	4.54	Strongly Agree

The system was evaluated by **3 Licensed IT Instructors / Practitioners**.**Table 12.4 Summary of ISO 25010 Software Quality Evaluation Results**

IT Instructors / Practitioners:

The IT instructors and practitioners who evaluate KUVENTORY are expected to assess its inventory management capabilities and recommend improvements to enhance its technical quality and operational effectiveness. Their feedback may focus on strengthening system security and data protection, particularly through secure hosting configurations and appropriate access controls. They may also recommend improving the search functionality for faster retrieval of inventory items and transaction records, resolving identified system errors, and continuously refining the graphical user interface (GUI) to provide a cleaner, more consistent, and user-friendly experience.

Persons Responsible for Evaluation

EVALUATOR GROUP	PERSON IN CHARGE	ROLE IN EVALUATION
Internal QA Testers	Melritch Jan Zipagan, Karla Ann DC, Calubaquib	Conducts functional testing, inventory transaction verification, database validation, and compatibility testing to identify and resolve system defects.
Faculty Evaluators	3 Licensed IT Instructors / Professors	Evaluates the system's technical implementation, functionality, performance, compatibility, and compliance with the selected ISO/IEC 25010 software quality criteria.

Table 12.5 Evaluation Stakeholders and Assigned Responsibilities

Recommendation

Based on the evaluation ratings and constructive feedback from the IT instructors, the following actions are recommended:

1. **Enhance GUI Design and Usability:** Improve the graphical user interface by maintaining a

consistent layout, readable typography, clear navigation, and uniform button styles across all modules. Ensure that inventory information, alerts, and reports are presented clearly to support efficient use.

2. **Strengthen System Security:** Reinforce authentication and role-based access control, review Supabase Row Level Security (RLS) policies, and apply secure hosting configurations and appropriate data protection measures to prevent unauthorized access to inventory records.

3. **Improve Frontend Performance and Responsiveness**: Optimize frontend components, scripts, and page rendering to ensure smooth navigation and responsive interactions. Verify that the system displays and functions properly across supported browsers, desktop computers, tablets, and mobile devices.

4. **Conduct Continuous Bug Fixing and Frontend Maintenance**: Regularly test the system to identify and resolve interface errors, broken components, display inconsistencies, and navigation issues. Refactor frontend code when necessary to improve maintainability, reliability, and consistency across modules.

13.0 APPENDICES

APPENDICES 1: INSTITUTIONAL PROFILE AND CLIENT BACKGROUND

A.1 Institutional Profile: Cagayan State University – Carig Campus

Cagayan State University – Carig Campus, situated in Tuguegarao City, Cagayan, serves as the university system's premier center for science, engineering, and technology education. Home to the College of Engineering and Architecture (COEA), the campus is dedicated to delivering high-quality tertiary education, promoting technical innovation, and fostering applied research across diverse engineering and computing disciplines. CSU Carig accommodates a dynamic student population, continually striving to modernize its instructional management and provide an academic environment that emphasizes technological competence, student accountability, and operational efficiency.

A.2 Client Background: Mrs. Feli Frogroso owner of Kape Uno Bistro

Kape Uno Restaurant, located in Carig Sur, Tuguegarao City, Cagayan, is the client establishment for KUVENTORY: Kape Uno Inventory Management System. The restaurant manages inventory across four operational stations: Bodega, Kiosk, Portion, and Grilled, each of which requires proper monitoring and recording of stock quantities and movements.

Before the development of KUVENTORY, inventory records were maintained manually using paper-based records and logbooks. This traditional process presented several operational challenges:

1. Time-Consuming Inventory Recording: Manually recording stock additions, deductions, and transfers across multiple stations required considerable time and effort, making it difficult to maintain updated inventory records.
2. Inventory Inaccuracy and Reconciliation Issues: Manual calculations and recording errors could result in discrepancies between recorded stock quantities and actual inventory, particularly during daily inventory reconciliation and shift changes.
3. Limited Stock Visibility and Coordination: The absence of centralized, real-time inventory monitoring made it difficult to determine current stock availability across Bodega, Kiosk, Portion, and Grilled, potentially causing shortages and delays in replenishing supplies.
4. Lack of Expiration and Low-Stock Monitoring: Without an automated system for tracking batch expiration dates and notifying personnel about low-stock items, perishable goods could expire before use, while necessary supplies might not be replenished promptly.

To address these challenges, KUVENTORY: Kape Uno Inventory Management System was proposed as a web-based solution designed to centralize inventory records, monitor stock quantities across the four stations, facilitate daily inventory reconciliation, implement First-Expired, First-Out (FEFO) batch tracking, and provide inventory alerts and reports. Through these functions, the system aims to improve inventory accuracy, strengthen stock movement traceability, and support more efficient inventory management at Kape Uno Restaurant.

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Documentation and Revision History

Revision Number	Date	Comment
1.0		

8.2 Local Environment Setup (Developer)

Table 8.2.1 Local environment setup (developer)

SETUP ACTIVITY	KAPEUNO DEVELOPER ACTION
Install project dependencies	Set up the KAPEUNO repository and required frontend development dependencies.
Configure Supabase	Provide the production/development project URL and public client configuration through safe environment variables.
Verify database schema	Compare frontend contracts with live tables, columns, relationships, constraints, functions, RPCs, and RLS.

SETUP ACTIVITY	KAPEUNO DEVELOPER ACTION
Prepare database state	Apply required migrations and safely remove confirmed test records without destroying schema or required configuration.
Run build checks	Resolve compilation, type, import, and configuration issues.
Prepare deployment	Confirm Netlify build settings, routes, assets, authentication redirect behavior, and production environment values.